MIT APPLIED AI AND DATA SCIENCE PROGRAM

Used-Cars Price Prediction for Cars4U

A Random Forest Regression Pricing Engine for the Indian Used-Car Market

Guillermo Daniel Bonilla
Capstone Project

April 12, 2026

PRESENTATION ROADMAP

Agenda

01

Overview of the Problem

Market instability, pricing ambiguity, and the Cars4U mandate

02

Approach for the Solution

Preprocessing pipeline, feature engineering, and model evaluation

03

Key Findings & Insights

EDA patterns, model comparison, and Random Forest performance

04

Recommendations & Next Steps

Deployment roadmap, stakeholder actions, and future enhancements

OVERVIEW OF THE PROBLEM

The Business Problem

Market Reality

The Indian used-car market is characterized by **price unpredictability**, **geographic disparities**, and **information asymmetry** between buyers and sellers.

Unlike new cars, used cars lack standardized pricing — leading to negotiation friction, inconsistent valuations, and eroded trust across the market.

✕ NO STANDARDIZED PRICING

VS

Cars4U's Mission

Cars4U must provide **accurate**, **unbiased**, and **scalable** price estimates that enhance clarity, reduce negotiation friction, and improve operational productivity.

The goal: develop a **machine-learning-driven pricing engine** that brings fairness, clarity, and scalability to every used car in inventory.

✓ ML-DRIVEN PRICING ENGINE

Bridging the gap with data-driven intelligence

OVERVIEW OF THE PROBLEM

Market Characteristics

Four defining data patterns that shape the pricing challenge

DIESEL DOMINANCE

53.1%

of used cars run on diesel fuel, reflecting 2010–2016 era fuel economics

FIRST-OWNER VEHICLES

82.1%

are first-owner cars, commanding higher resale prices and buyer trust

PRICE SKEWNESS

Heavy Right-Skew

Luxury vehicles create long right-tails; median ₹5.64L vs a much higher mean of ₹9.48L (values in INR 100K)

MISSING DATA

6,246

missing New Price values out of 7,253 rows — the largest data quality challenge

SOURCE: Cars4U Dataset — 7,253 listings across 10 Indian cities

APPROACH FOR THE SOLUTION

Data & Preprocessing Pipeline

From raw dataset (7,253 rows) to model-ready features — a systematic 4-step transformation

1 Data Cleaning

Dropped Serial Number (ID field) – not useful for modeling

Handled missing values via median imputation strategies and dropping values:
New Price: Three-level hierarchical fill – Brand-Model median → Brand median → Overall median

Seats: Two-level hierarchical fill – Brand-Model median → Overall median
Mileage, Engine, Power: Median imputation for each variable
Removed 1,234 rows with missing Price/Price log values needed for modeling

Missing values count:

- New Price 6,246
- Price 1,234
- Log(Price) 1,234
- Power 175
- Mileage 83
- Seats 53
- Engine 46

2 Outlier Treatment

Addressed the heavy right-skew caused by not capping or removing outliers
Extreme values in numerical variables reflect real used-car characteristics – especially premium vehicles – so keeping them preserves true variance and supports stronger model performance

Outliers found in:

- Price
- Kilometers Driven
- Power
- Engine
- Mileage
- New Price
- Car Age
- Depreciation Amount

3 Log Transformations

Normalized distributions, improved linearity, reduced outlier influence

Log transformed variables:

- Log(Price) target
- Log(Kilometers Driven)

skewed → normal

4 Encoding and Rare Category Grouping

One-hot encoded categorical variables and Seats discrete variable
Grouped rare categories (Fuel Type, Owner Type, Seats, Brand) to stabilize models and avoid sparse-category issues

Encoded categorical variables count:

- Location 10
- Brand 16
- Seats 4
- Owner Type 2
- Fuel Type 2
- Transmission 1

43 FEATURES

4,212 TRAIN SAMPLES

Pipeline output: 7,253 raw rows → cleaned, outlier-treated, log-transformed, and one-hot encoded into **43 features** across **4,212 training samples** ready for model evaluation.

APPROACH FOR THE SOLUTION

Feature Engineering

Car Age

Computed as reference year (2019) minus manufacturing Year. Captures depreciation dynamics — older vehicles exhibit predictable value decline patterns.

Pearson r = −0.50 with Log(Price)

Depreciation Amount

Calculated as New Price minus current Price. Strongly correlated with Car Age (r=0.20) and Mileage (r=0.24); captures brand- and model-level value loss over time.

Brand & model value-loss signal

Brand and Model Extraction

Both extracted from the Name variable. Luxury brands (Lamborghini, Bentley, Porsche) show depreciation of −108 to −7 INR 100K vs mass-market brands at −5 to −1 INR 100K. In total, there are 218 models; used to impute Seats and New Price.

16 brands encoded across segments

Log(Price) Target and Log(Kilometers Driven)

Log-transformed Price variable; reduces extreme high-value distortions from luxury vehicles and produces a more stable, model-friendly target distribution.
Log-transformed Kilometers Driven variable; compresses extreme values up to 800,000 km into a model-friendly scale, reducing outlier influence on predictions.

Pearson r = −0.21 Log(Kilometers Driven) with Log(Price)

APPROACH FOR THE SOLUTION

Train 70 / Test 30

Model Evaluation Framework

Six regression models evaluated with hyperparameter tuning across five performance metrics

1 Multiple Linear Regression
Baseline linear model assuming additive feature effects

2 Ridge Regression
L2-regularized model that shrinks large coefficients to reduce overfitting

3 Lasso Regression
L1-regularized model that drives unimportant coefficients to zero

4 KNN Regression
Non-parametric model predicting from the nearest data points

5 Decision Tree Regression
Nonlinear model using recursive splits (tree-based splits) to capture interactions

Random Forest Regression FINAL CHOICE

Ensemble of decision trees that improves stability and generalization – superior accuracy and generalization

EVALUATION METRICS

R^2 | Adj. R^2 | RMSE | MAE | MAPE

Model Evaluation Approach
Models were evaluated consistently using train/test predictions, with the log transform on the Price variable reversed as needed. Performance was assessed using R^2/Adjusted R^2 for explained variance and RMSE, MAE, and MAPE for error metrics, providing a clear view of fit and generalization.

Key Insight
Linear models (1–3) struggled with nonlinear feature interactions. Tree-based models (5–6) captured complex pricing patterns, with **Random Forest** delivering the best generalization through ensemble averaging. KNN Regression (non-parametric) performed moderately well – better than linear models but below tree-based models.

KEY FINDINGS & INSIGHTS

EDA Insights — Pricing Drivers

Pearson correlation analysis reveals the key economic relationships driving used-car resale value.

Power vs Log(Price) r = 0.76

Higher horsepower drives strongly increases resale value – the single most predictive feature.

Engine Size vs Log(Price) r = 0.68

Larger engines command higher prices across fuel types and segments.

New Price vs Log(Price) r = 0.32

Originally expensive cars retain more value – brand premium persists in resale.

Car Age vs Log(Price) r = −0.50

Clear negative relationship — predictable depreciation year over year.

Log(Kilometers Driven) vs Log(Price) r = −0.21

More usage lowers value, though moderately – age has a stronger effect than mileage.

Location/Geographic Variability Regional

Coimbatore and Bangalore show wide price dispersion; Jaipur and Kochi show tighter, more stable pricing.

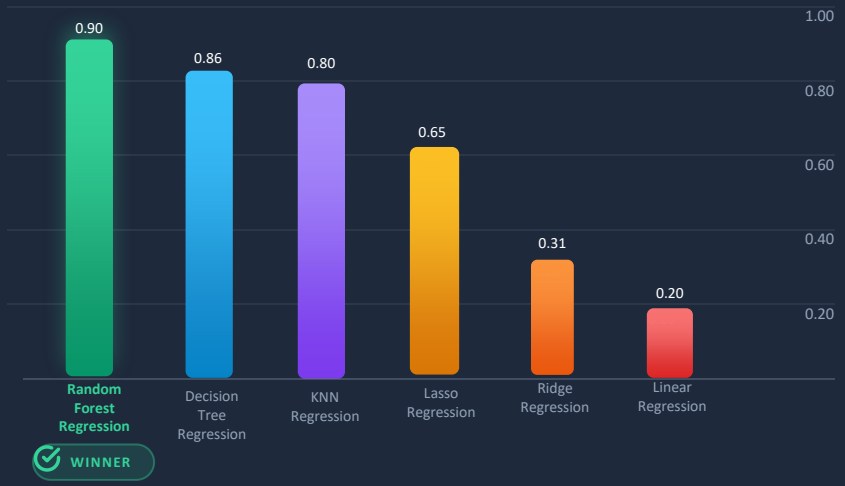
- Wide: **Coimbatore, Bangalore**
- Tight: **Jaipur, Kochi**

Bar lengths represent absolute |r| values — visual scale of correlation strength

KEY FINDINGS & INSIGHTS

Model Performance Comparison

TEST R^2 SCORE (HIGHER IS BETTER)



FULL TEST METRICS

MODEL	R^2	RMSE	MAE	MAPE
Linear Regression	0.20	9.97	2.11	18%
Ridge Regression	0.31	9.26	2.01	18%
Lasso Regression	0.65	6.61	2.91	28%
KNN Regression	0.80	5.04	2.29	25%
Decision Tree Regression	0.86	4.20	1.21	13%
Random Forest Regression	**0.90**	**3.59**	**0.88**	**9%**

Linear models failed to capture nonlinear feature interactions ($R^2 < 0.70$). **Tree-based models excelled** — Random Forest achieved **4.5× higher R^2** and 94.1% lower RMSE than linear regression, confirming that complex pricing dynamics require ensemble methods. **KNN Regression** performed adequately, outperforming linear models but falling short of tree-based models.

Linear Models · Non-parametric · Tree-based

RMSE in INR 100K · Train/Test: 70/30

KEY FINDINGS & INSIGHTS

Random Forest Regression — Final Model

Champion model delivering production-grade pricing accuracy across the Indian used-car market

BEST

0.90

Test R-squared

Explains 90% of price variance on unseen data

LOWEST

3.59

Test RMSE

Lowest average prediction error across all models (INR 100K)

BEST

9%

Test MAPE

Smallest percentage error — highest pricing accuracy

BEST

0.97

Train R-squared

Strong fit with modest train-test gap indicating healthy generalization

MODEL CONFIG

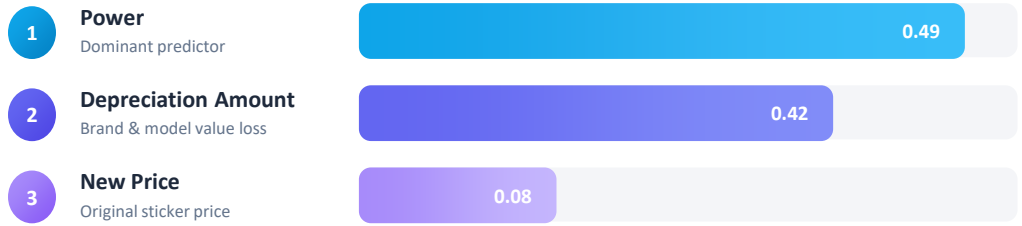
max_depth	5
min_samples_split	2
min_samples_leaf	1
n_estimators	100
bootstrap	True
random_state	1

KEY FINDINGS & INSIGHTS

Feature Importance — What Drives Pricing

Decision Tree | Random Forest

RANDOM FOREST FEATURE IMPORTANCE SCORE (Relatively the same as Decision Tree)



Top 2 features account for **91%** of total model importance

KEY INSIGHT

Power and **Depreciation Amount** together dominate the model's predictive signal, confirming that *engine performance* and *value retention* are the primary economic drivers of used-car pricing in India.

Power (HP)

Highest importance across both Decision Tree and Random Forest — horsepower is the strongest single price signal.

Depreciation Amount

Captures brand- and model-level value loss — an engineered feature that outperforms raw New Price.

New Price

Original sticker price influences resale value – but is far less predictive than performance and depreciation.

RECOMMENDATIONS & NEXT STEPS

Implementation Recommendations

01

Integrate into Pricing Pipeline

Embed the Random Forest model into seller onboarding so every new listing receives an automated, data-driven price. Use predictions to flag overpriced and underpriced vehicles in real time.

02

Deploy Feature-Importance Dashboards

Use SHAP and permutation importance to provide sellers with transparent pricing explanations. Build a buyer-facing UI showing which factors most impact each car's valuation.

03

Establish Retraining Cycles

Set up weekly or monthly data refresh and model retraining pipelines to prevent model drift as market conditions, fuel prices, and consumer preferences evolve over time.

04

Expand Data Sources

Incorporate service history, accident records, seller reputation scores, and real-time market demand indicators to boost predictive capability and capture latent pricing signals.

Each recommendation builds on the Random Forest model's Test $R^2 = 0.90$ performance to maximize business impact

RECOMMENDATIONS & NEXT STEPS

Stakeholder Roadmap & Next Steps

DEPLOY

Data Science Team

Deploy Random Forest into production, automate retraining pipelines, and maintain feature-importance dashboards.

Engineering / Platform Team

Build the inference pipeline for batch and real-time predictions, implement the pricing API, and design monitoring and alerting systems.

Product & Business Teams

Integrate model-generated prices into the UI, develop customer-facing explanations, and pilot geo-based pricing strategies.

Leadership

Approve deployment budgets, define pricing governance, and review monthly accuracy KPIs.

Future Model Enhancements

Apply Bayesian hyperparameter tuning, benchmark against XGBoost/LightGBM/CatBoost, add interaction features, and build a depreciation-forecasting module.

EVOLVE

CLOSING

Thank You

It has been a privilege to present this capstone project. I welcome your questions, feedback, and discussion.

Key Takeaways

1. The Indian used-car market requires a **machine-learning pricing engine** to reduce **price instability**, information asymmetry, and negotiation friction.

2. The **Random Forest Regression** model outperformed all five alternatives, achieving **R^2 = 0.90**, **RMSE = 3.59**, and **MAPE = 9%** on unseen data.

3. **Power** and **Depreciation Amount** dominate the model's predictive signal, aligning with economic intuition about **engine performance** and **value retention**.

4. Deploying this model can **reduce mispricing errors by 20–30%**, accelerate inventory turnover, and **strengthen buyer-seller trust** for Cars4U.

Appendix (Important Visualizations and Tables)

Figure A1. Price (INR 100,000) Distribution by Brand (With Outliers)

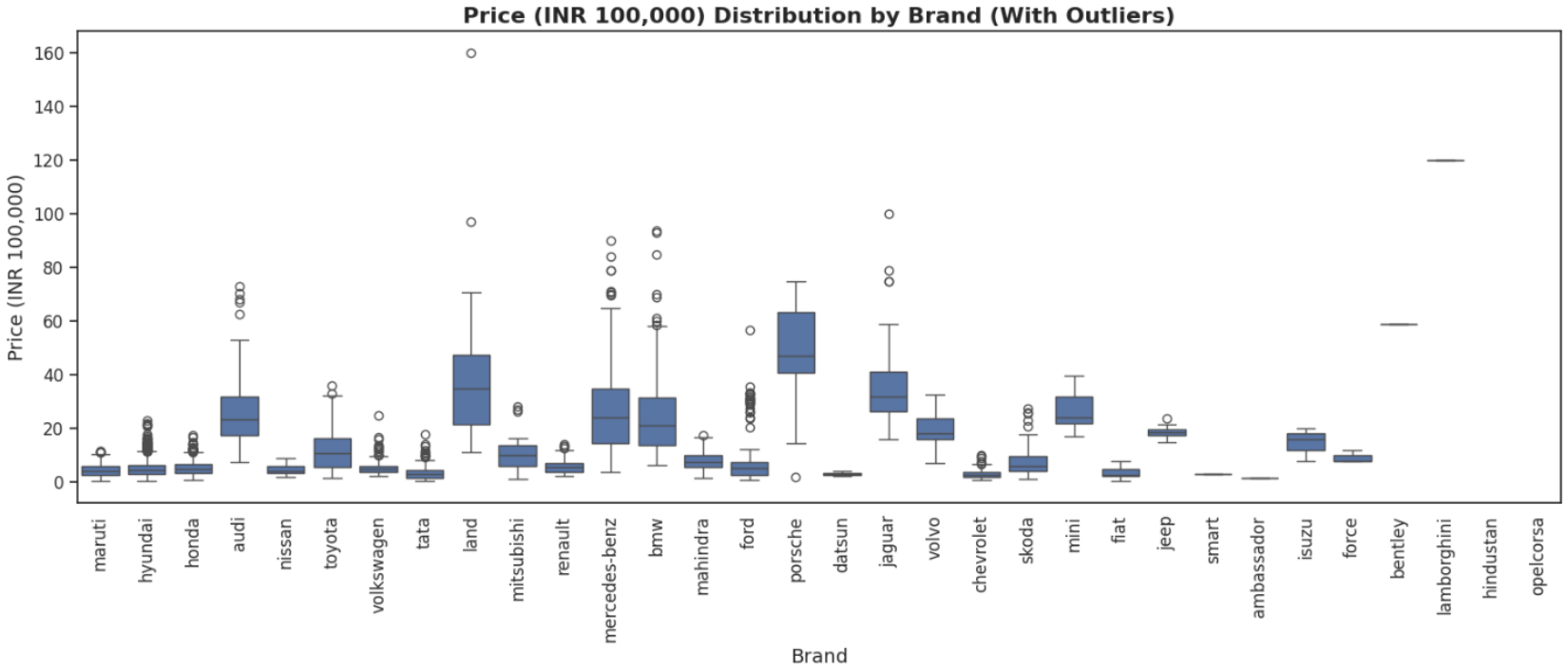


Figure A2. Price (INR 100,000) Distribution by Location (With Outliers)

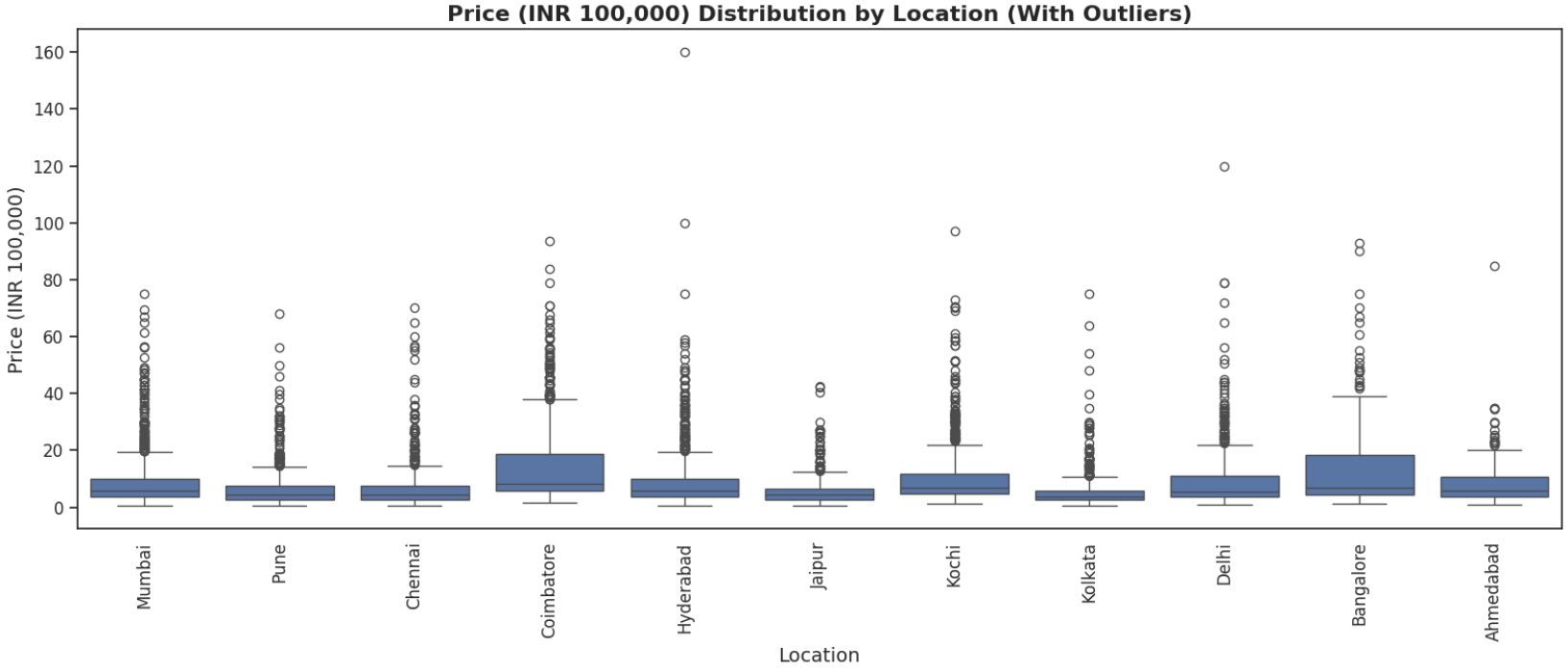


Figure A3. Vehicle Counts by Fuel Type

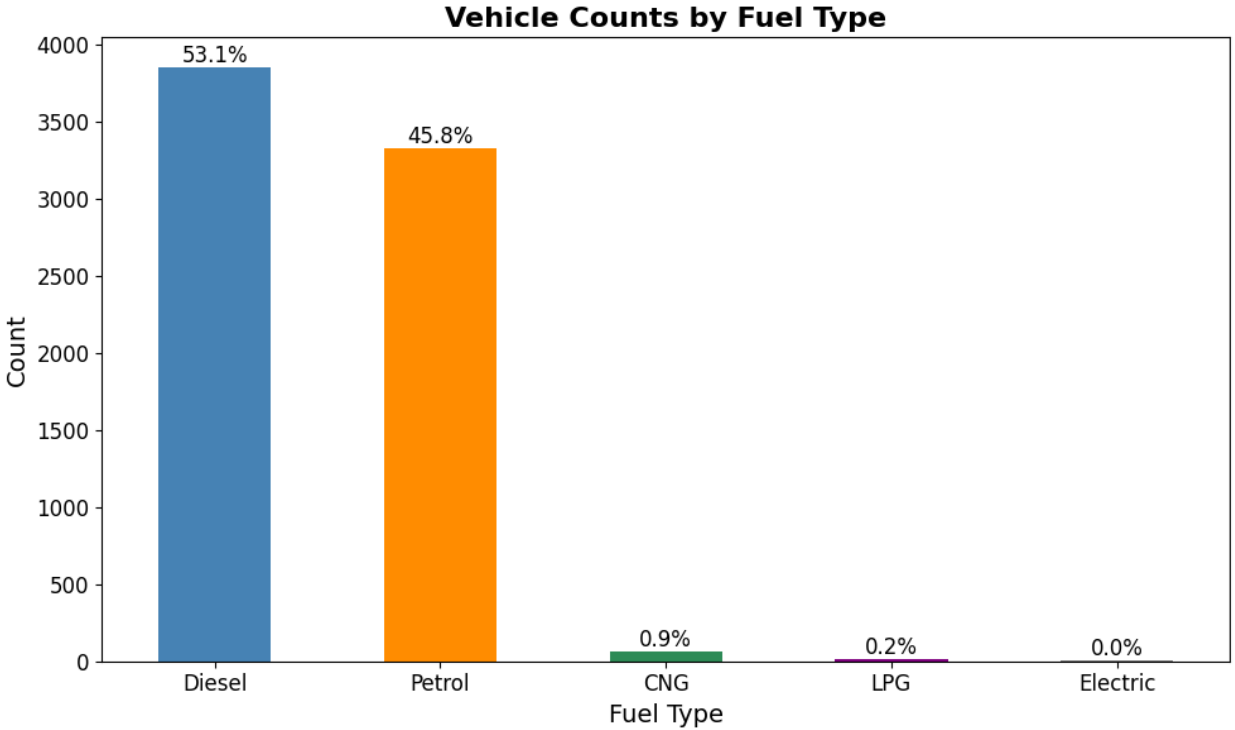


Figure A4. Vehicle Counts by Owner Type

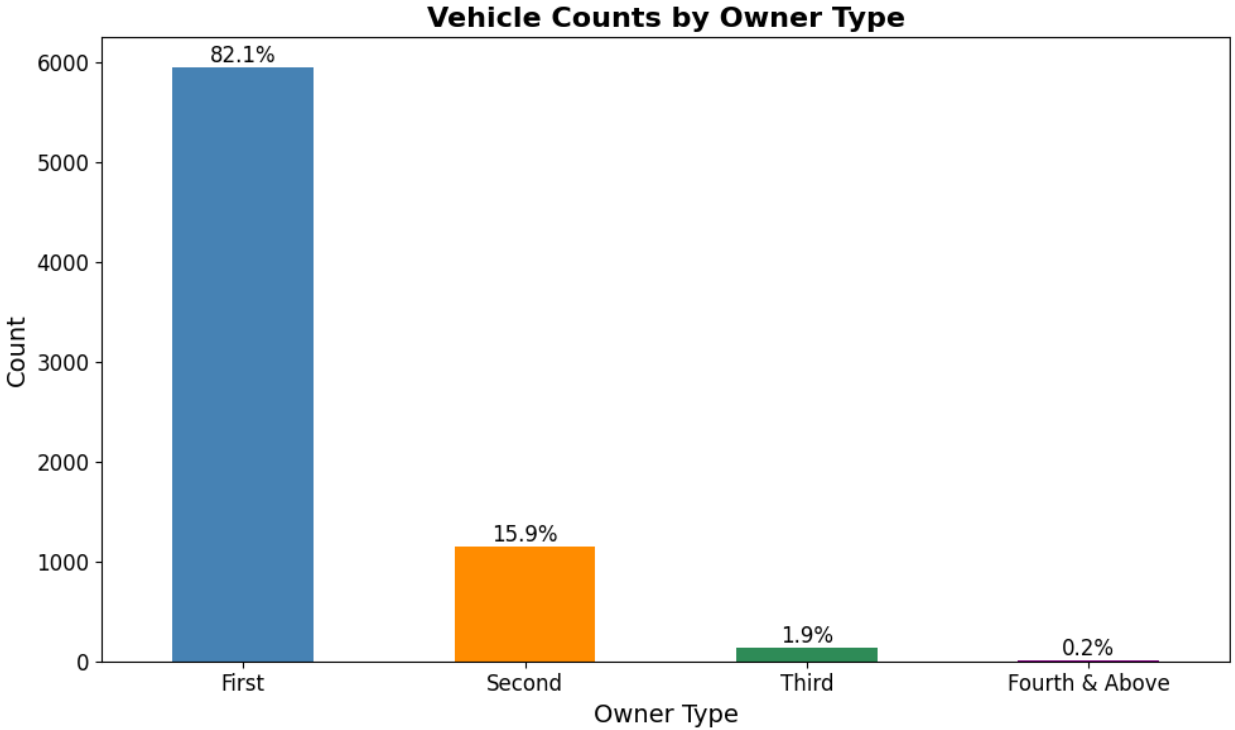


Figure A5. Price (INR 100,000): Boxplot and Histogram

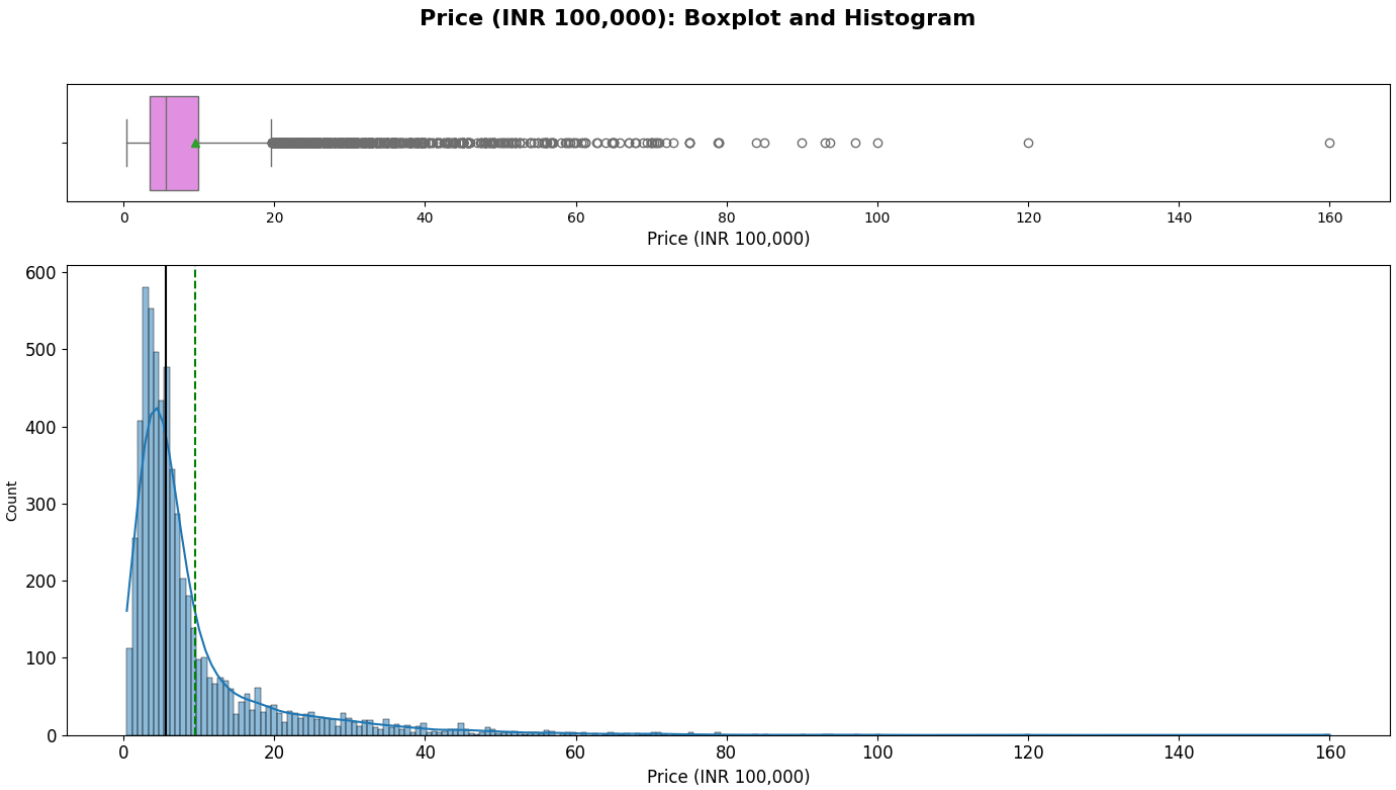


Figure A6. Kilometers Driven (KM): Boxplot and Histogram

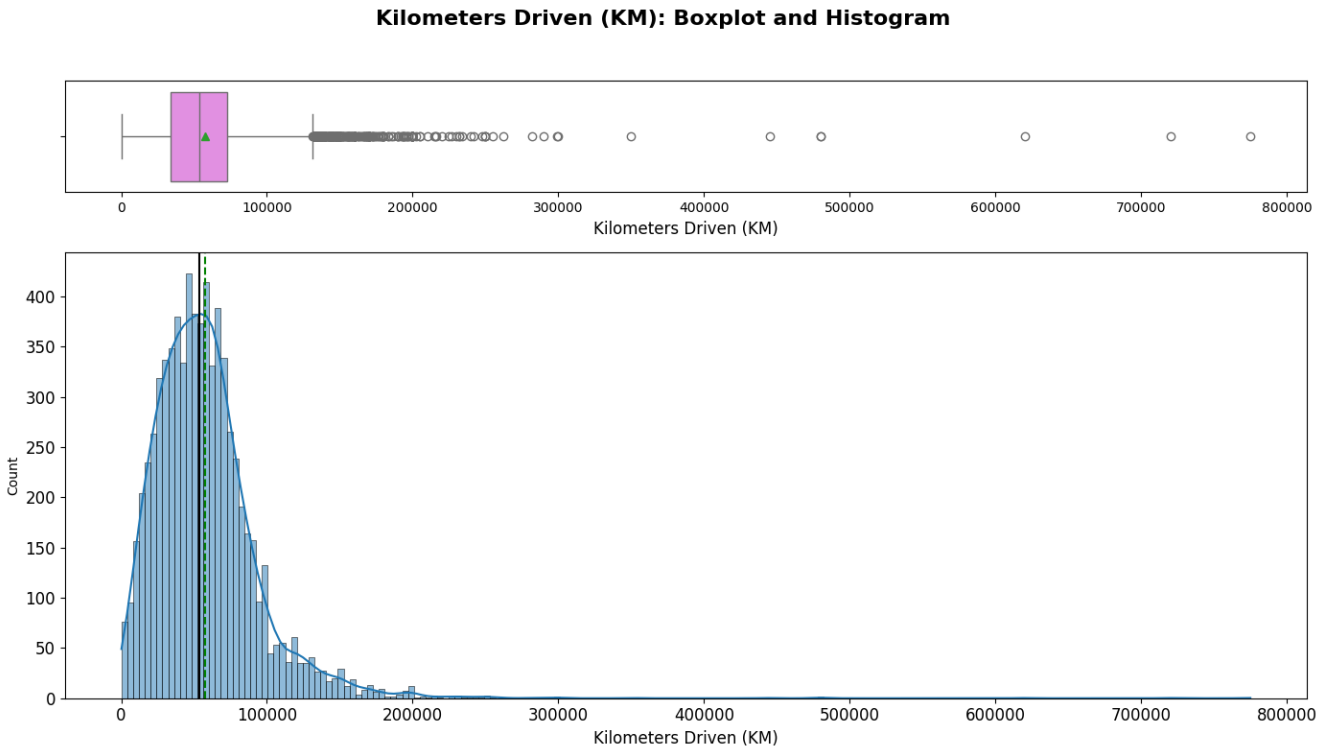


Figure A7. Power (BHP): Boxplot and Histogram

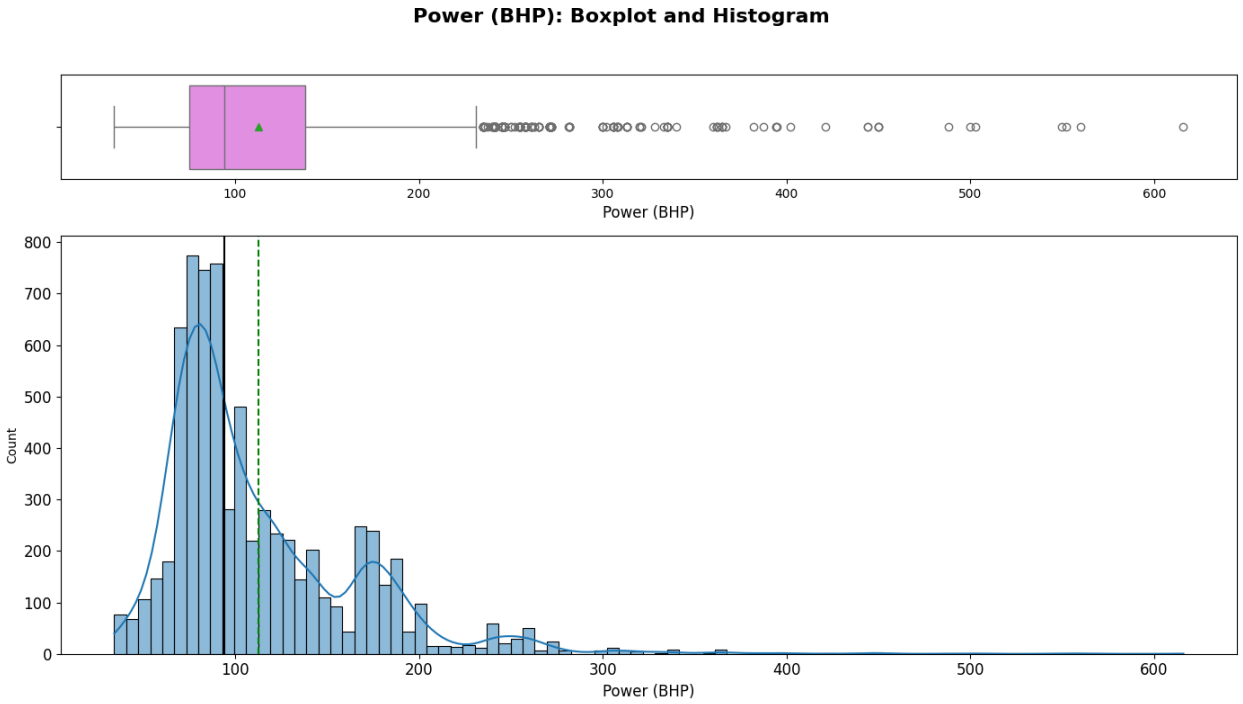


Figure A8. New Price (INR 100,000): Boxplot and

Figure A9. Engine (CC): Boxplot and Histogram

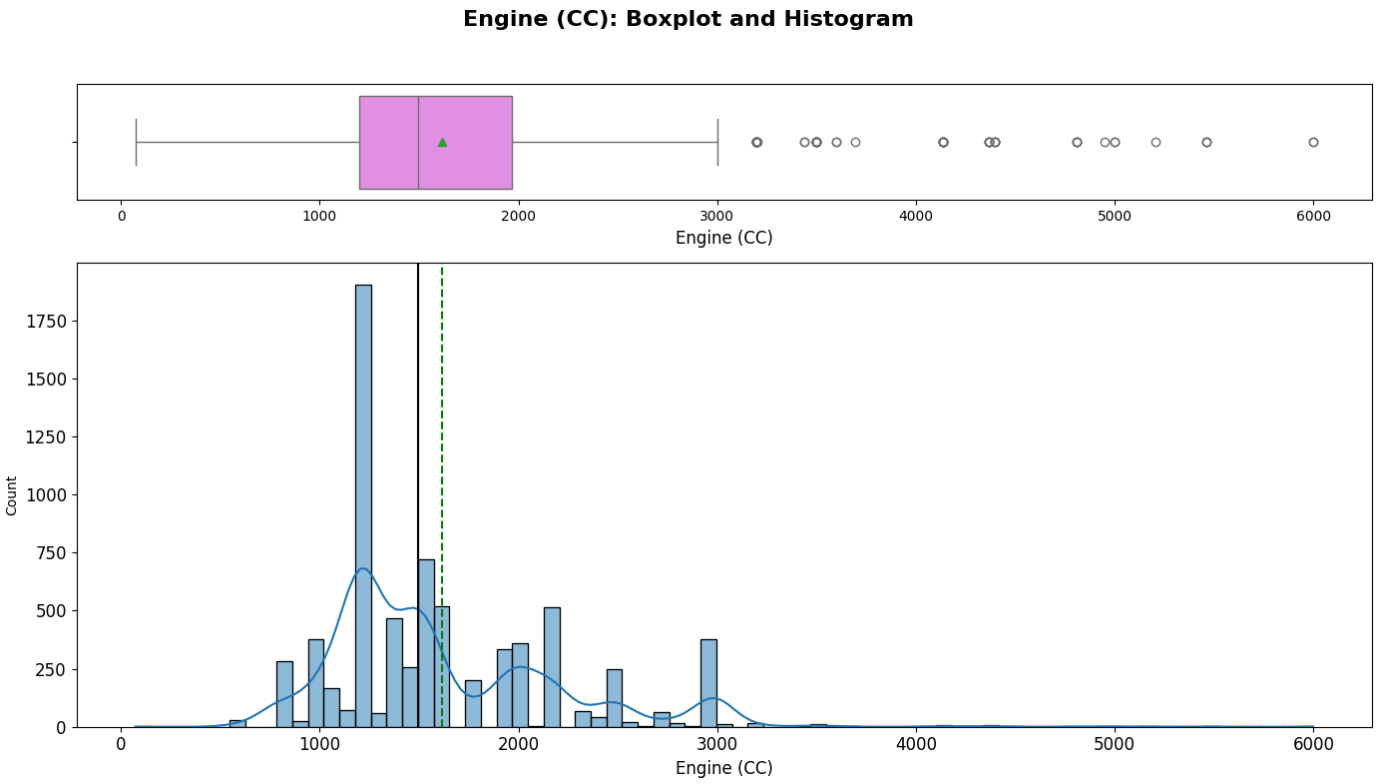


Figure A10. Mileage (KMPL or KM/KG): Boxplot and Histogram

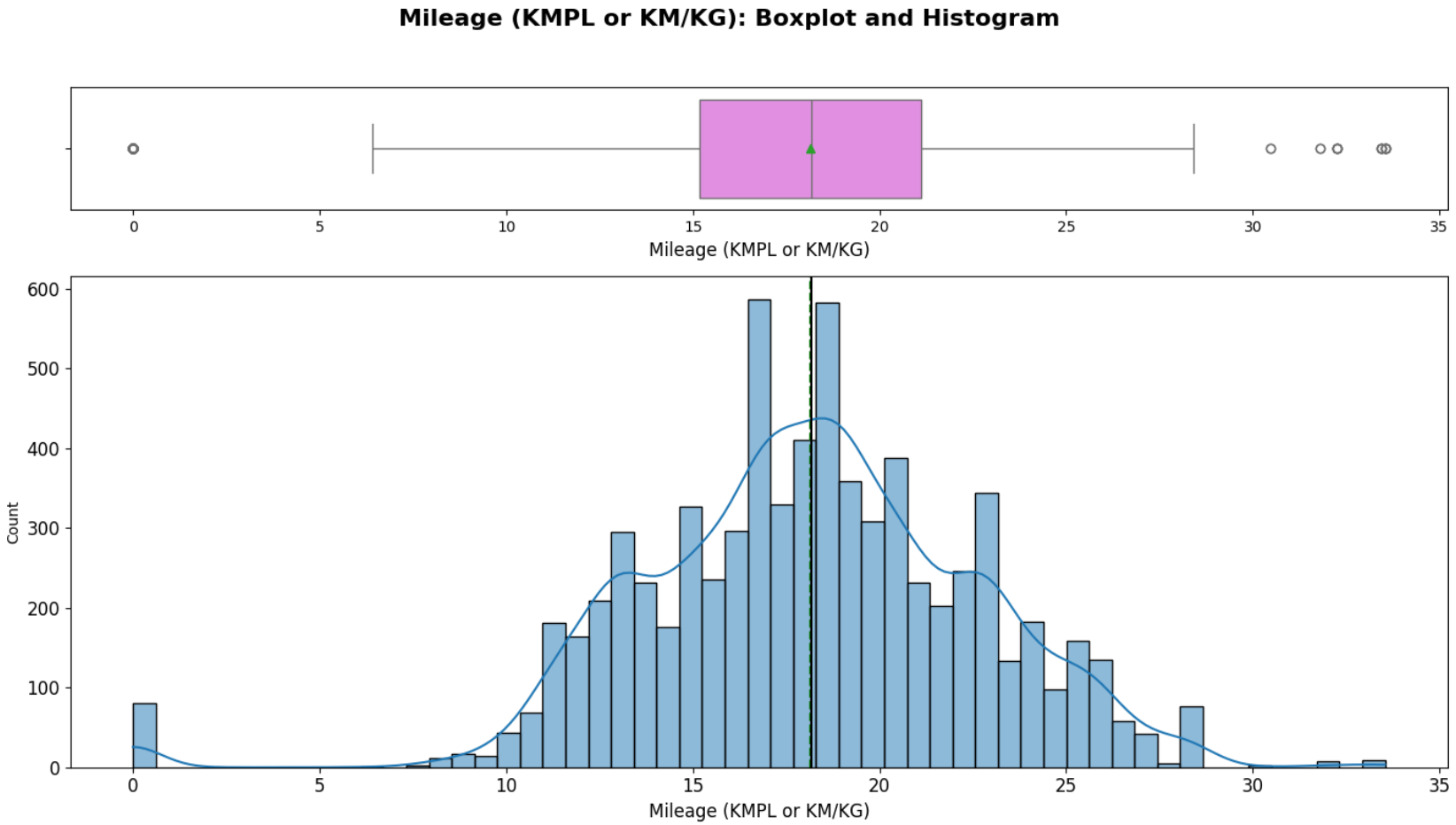


Figure A11. Variables Ranked by Missing Data

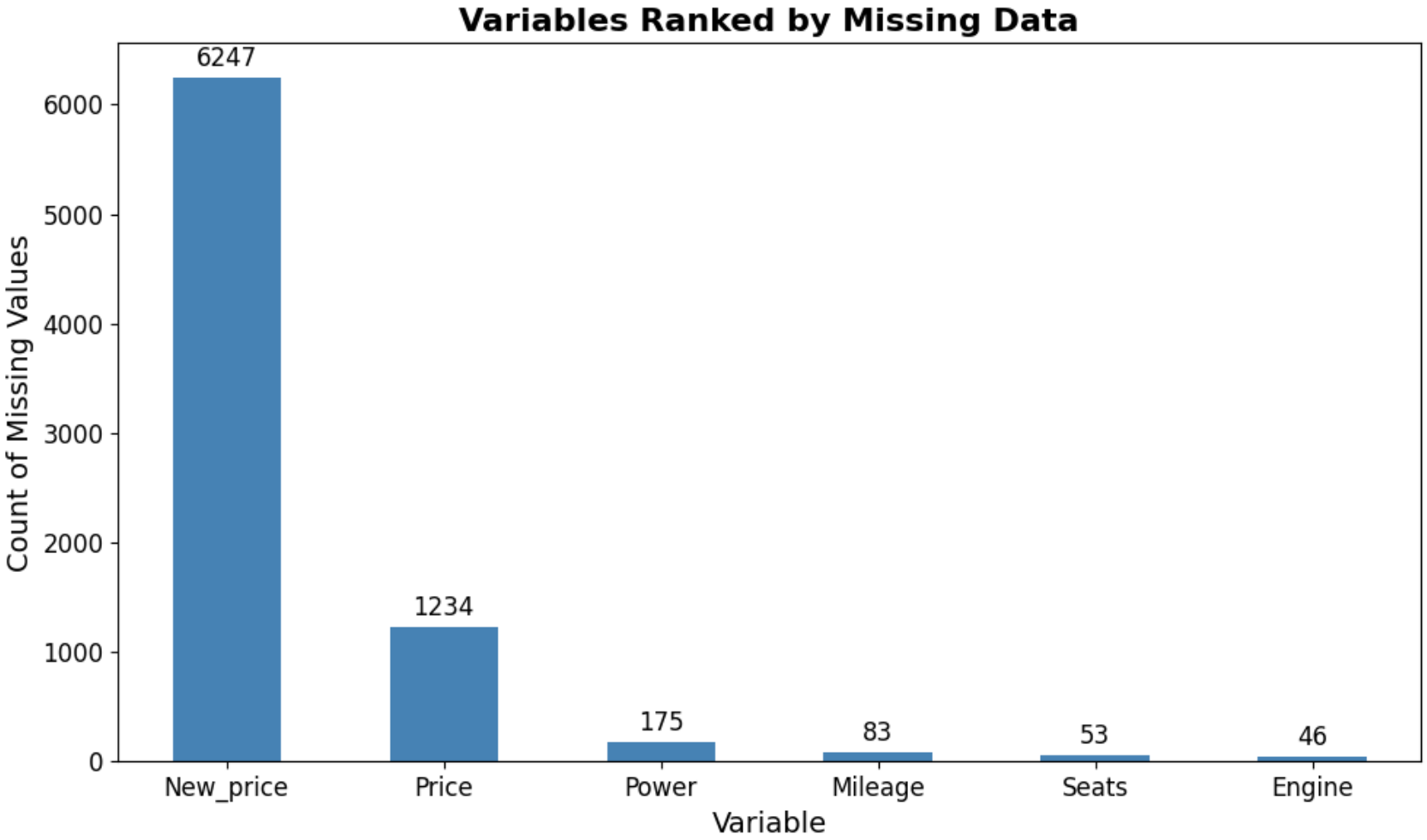


Figure A12. Correlation Matrix of Key Predictors

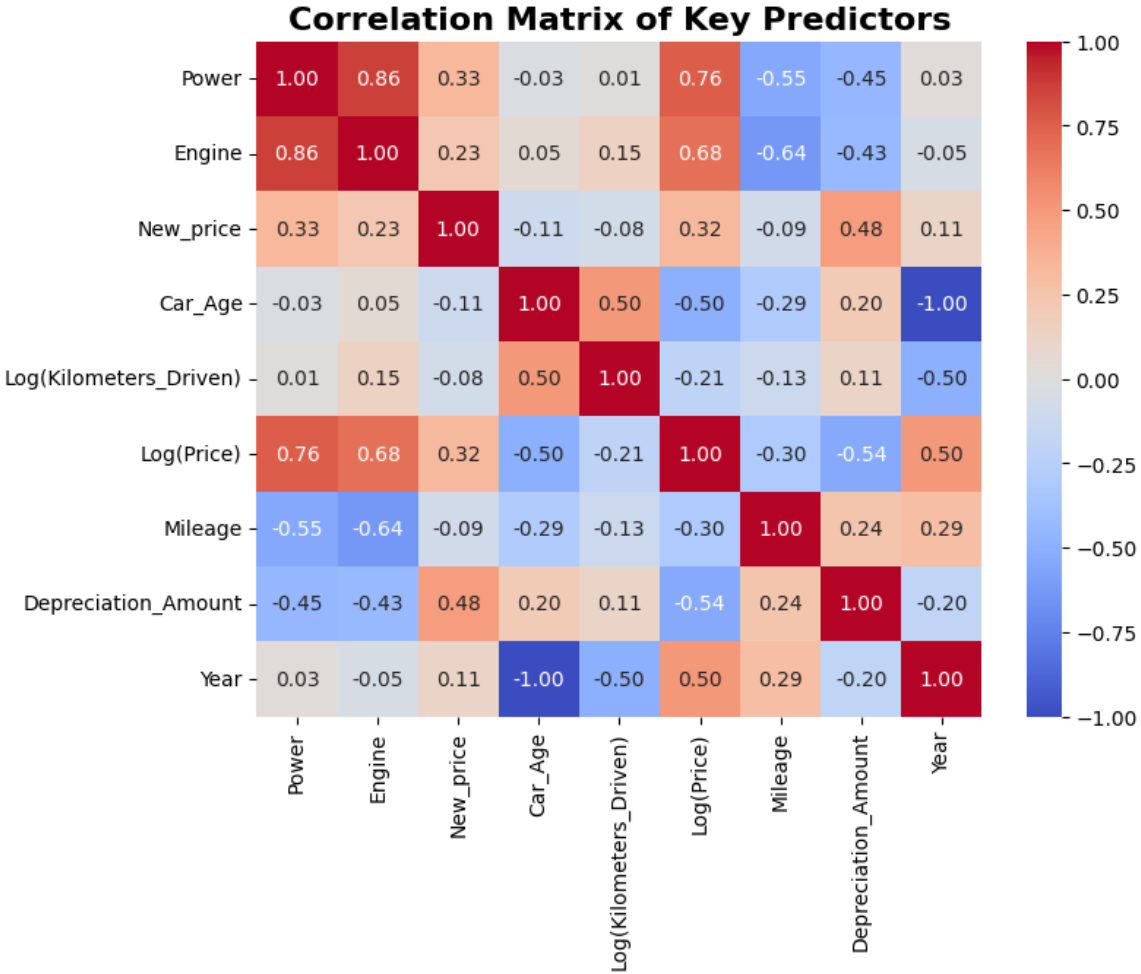


Figure A13. Power (BHP) vs. Log(Price)

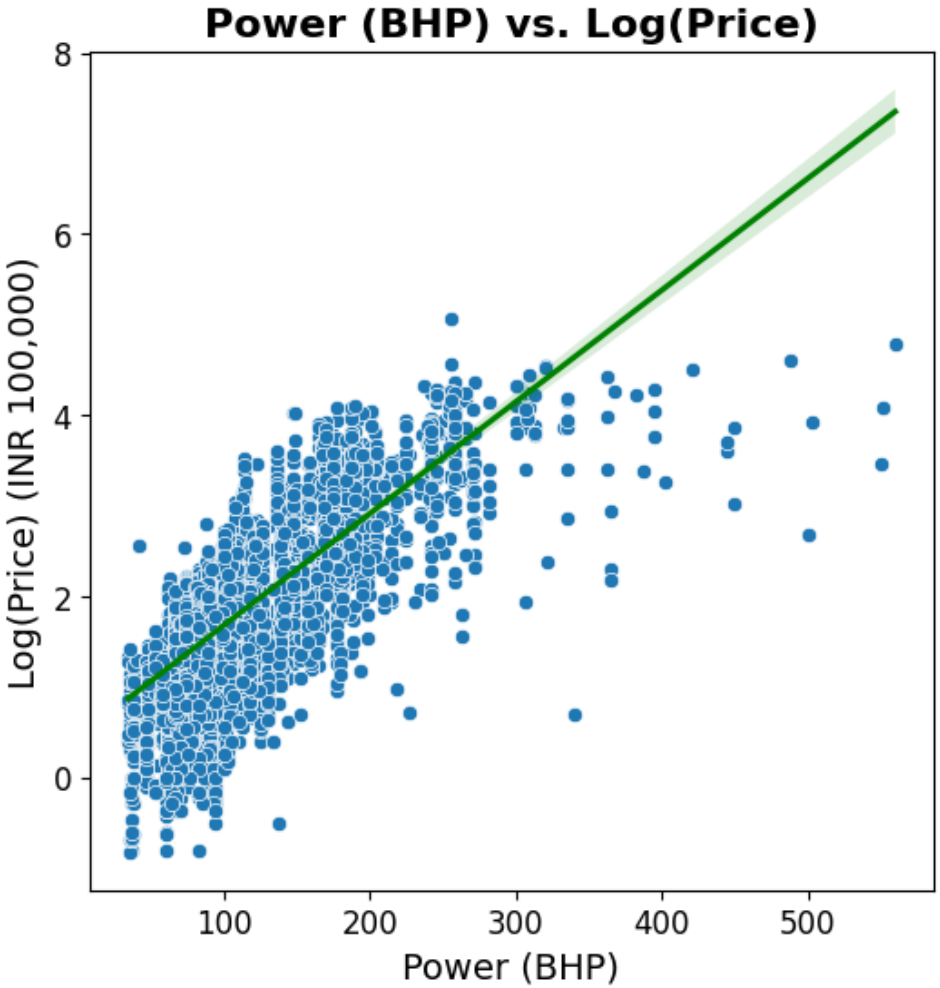


Figure A14. New Price vs. Log(Price)

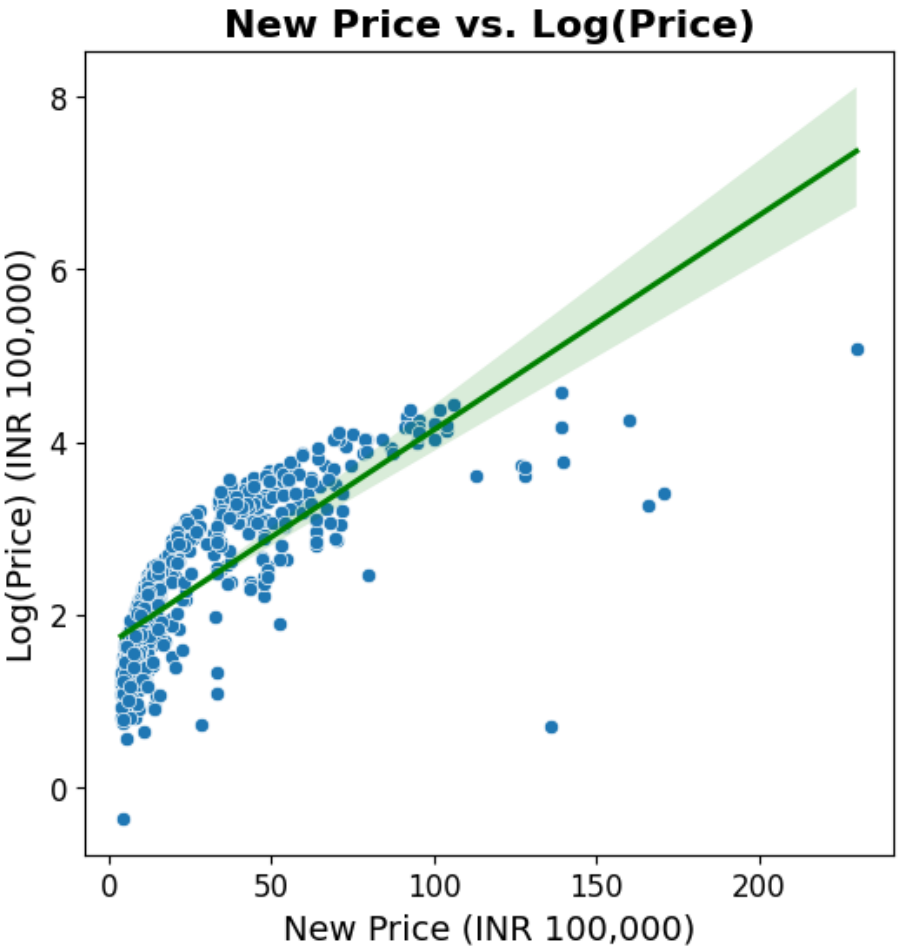


Figure A15. Engine (CC) vs. Log(Price)

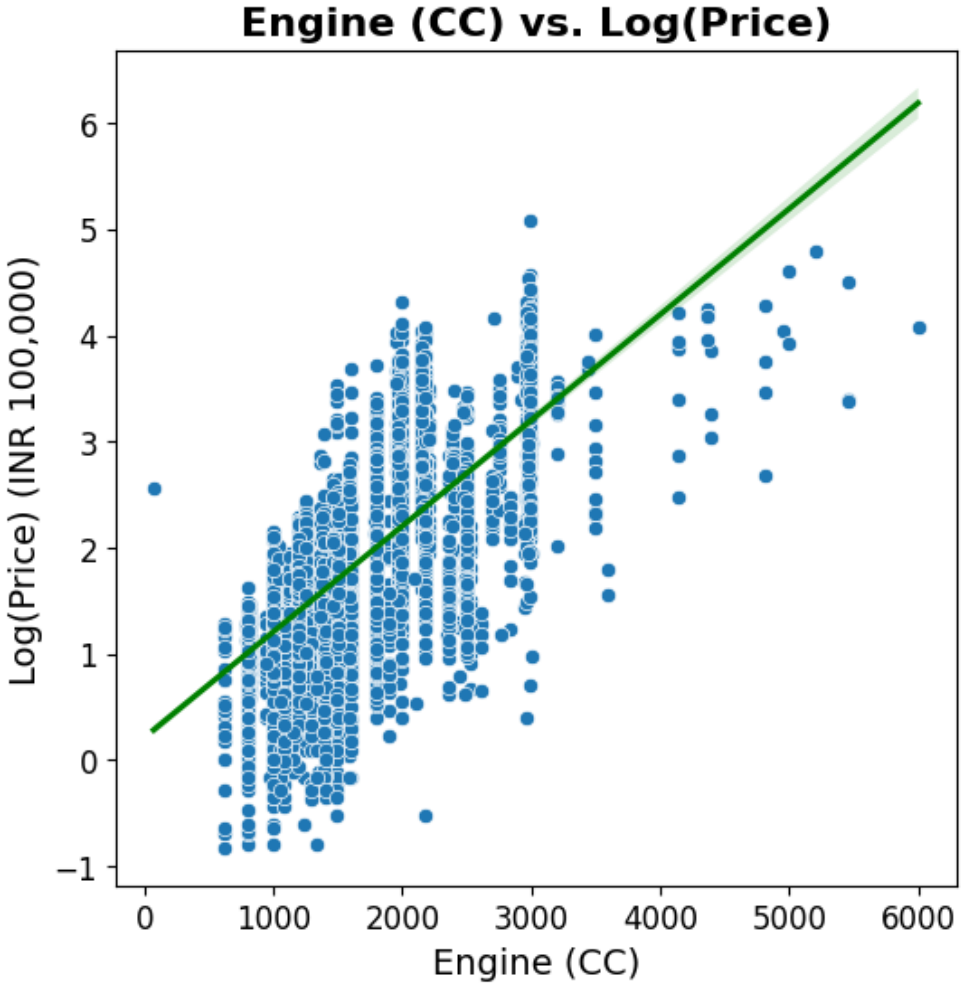


Figure A16. Manufacturing Year vs. Log(Price)

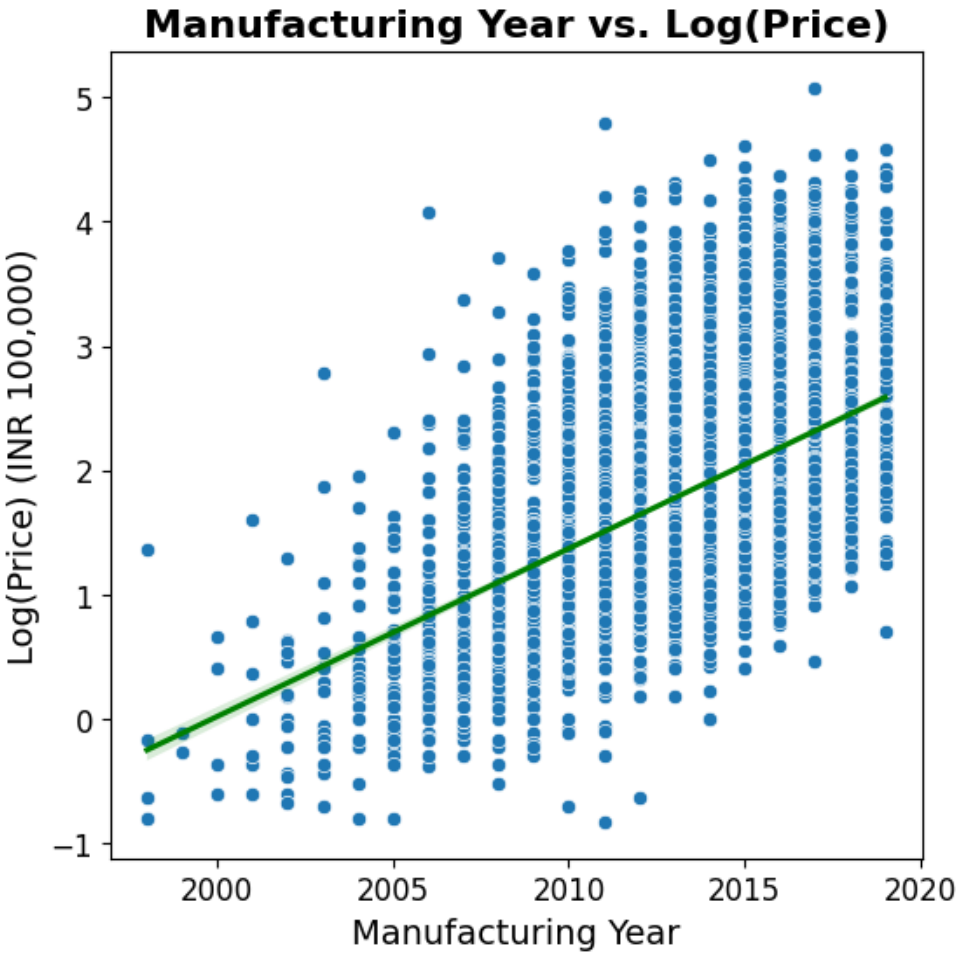


Figure A17. Car Age vs. Log(Price)

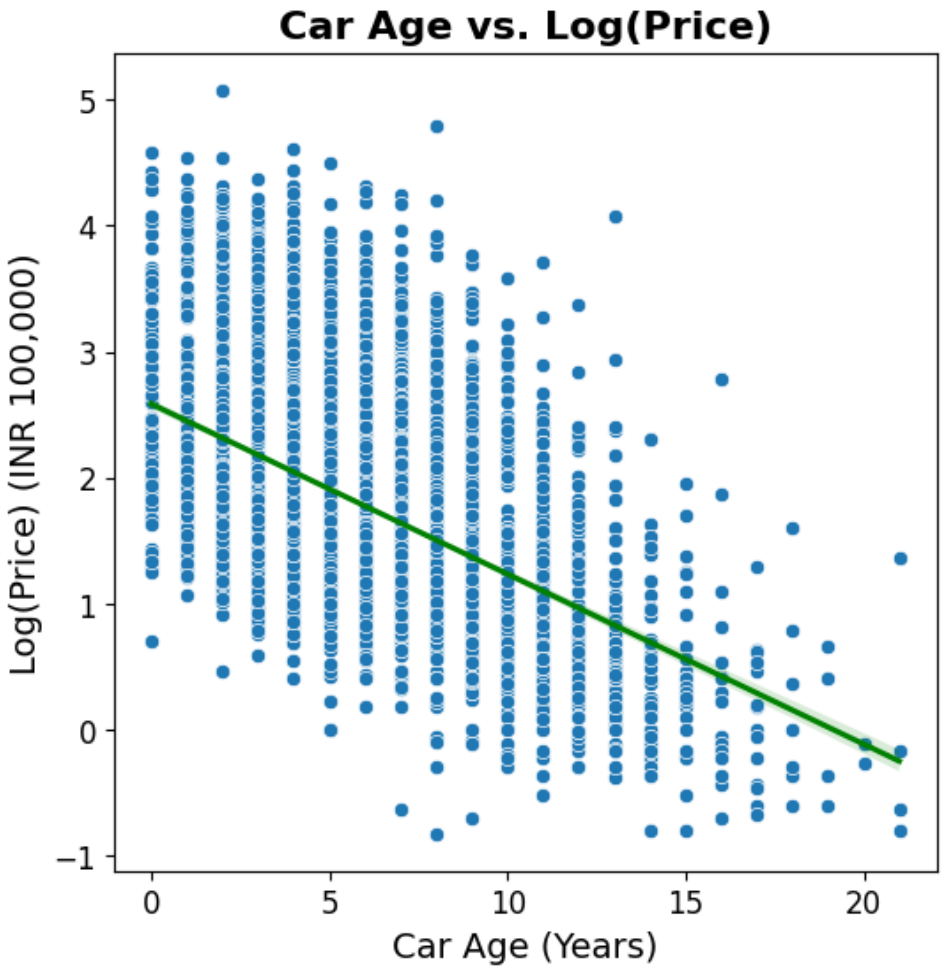


Figure A18. Log(Kilometers Driven) vs. Log(Price)

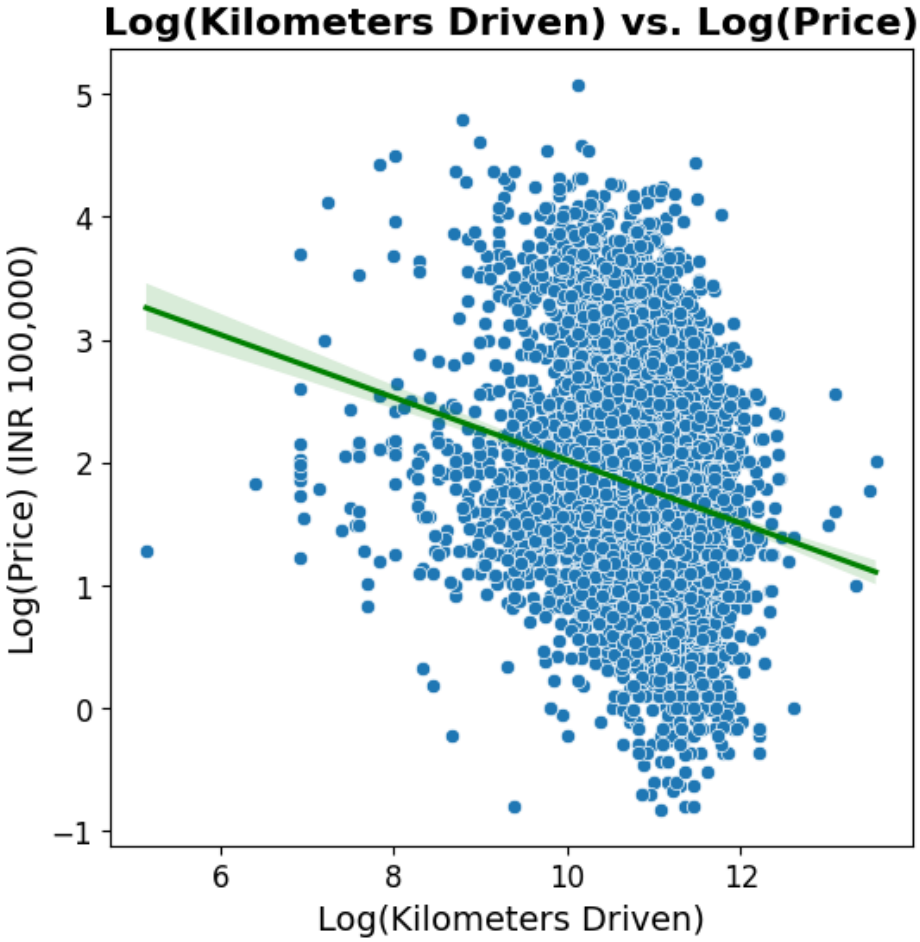


Figure A19. Mileage (KMPL or KM/KG) vs. Log(Price)

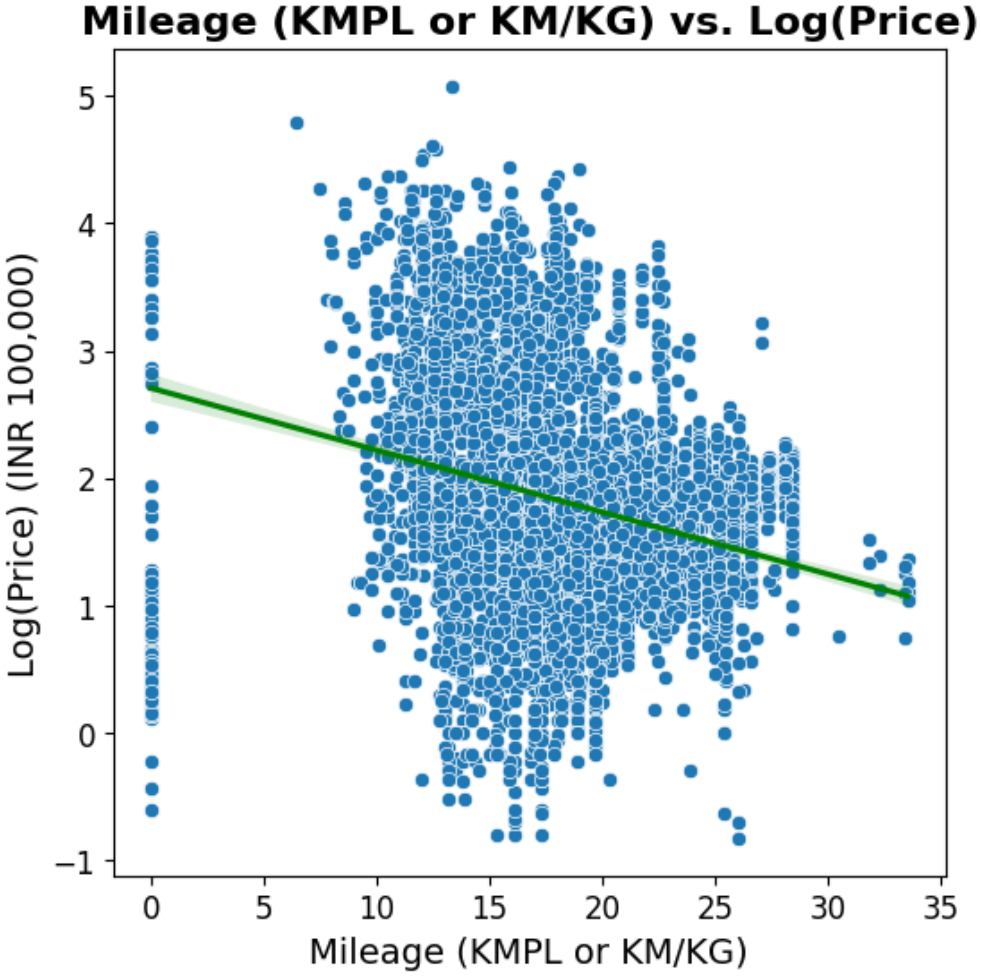
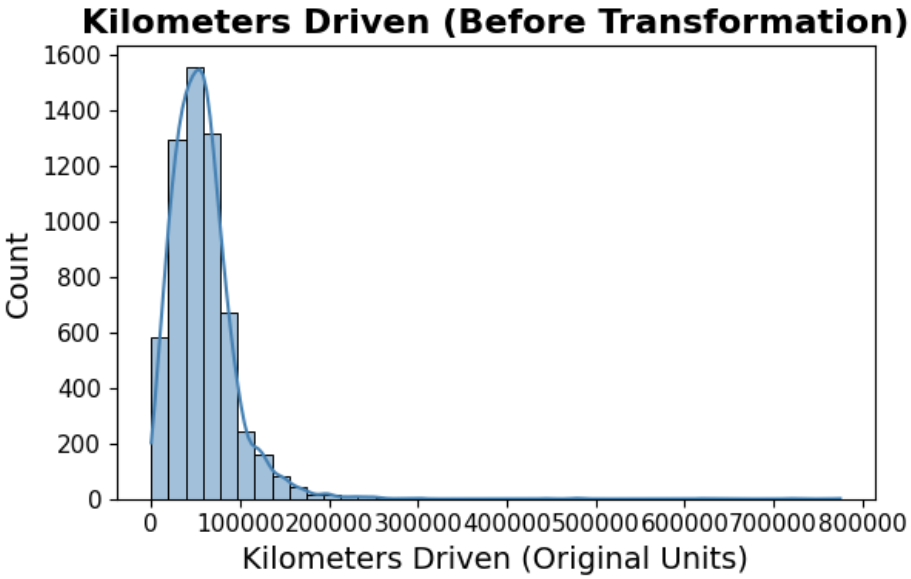


Figure A20. Kilometers Driven (Before and After Log Transformation)



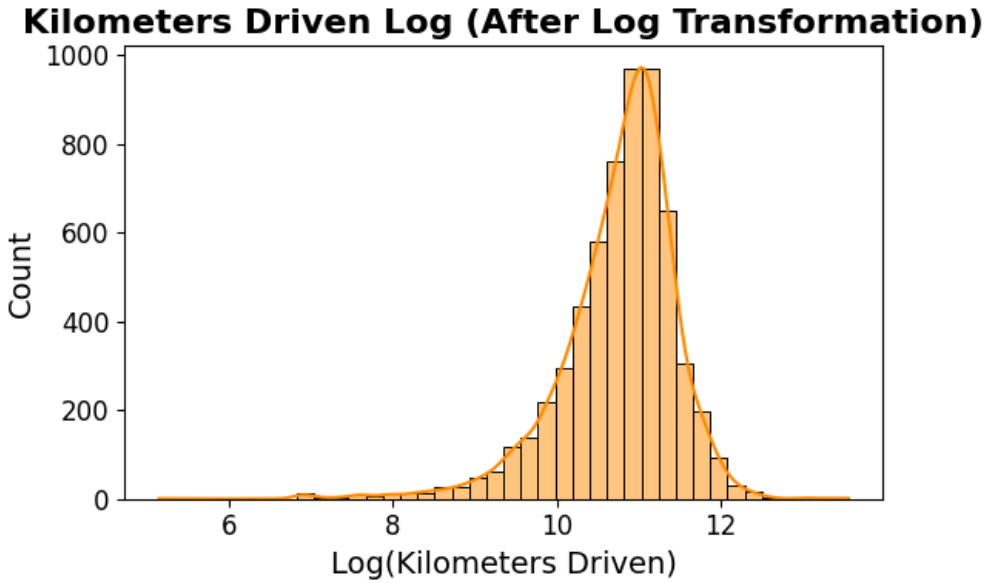
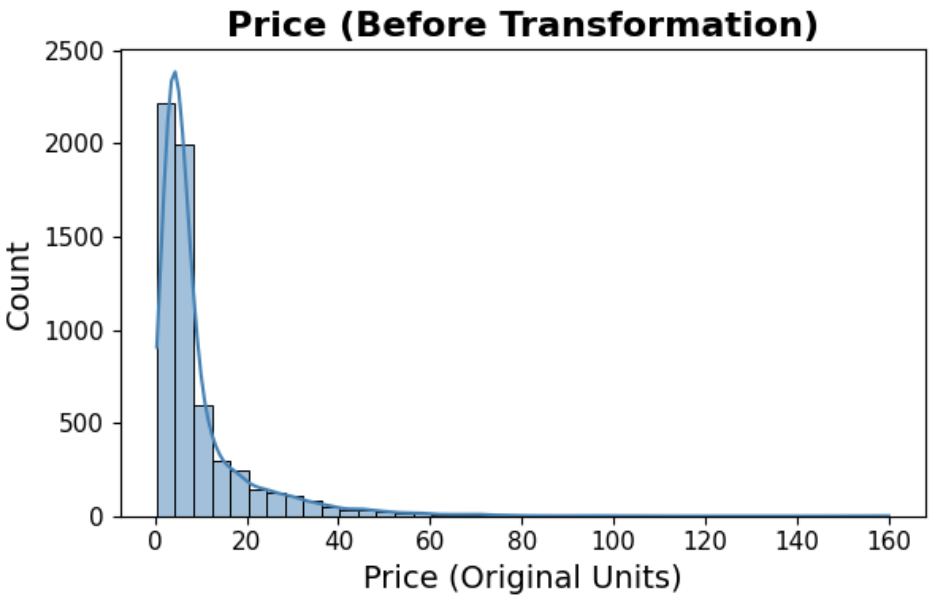


Figure A21. Price (Before and After Log Transformation)



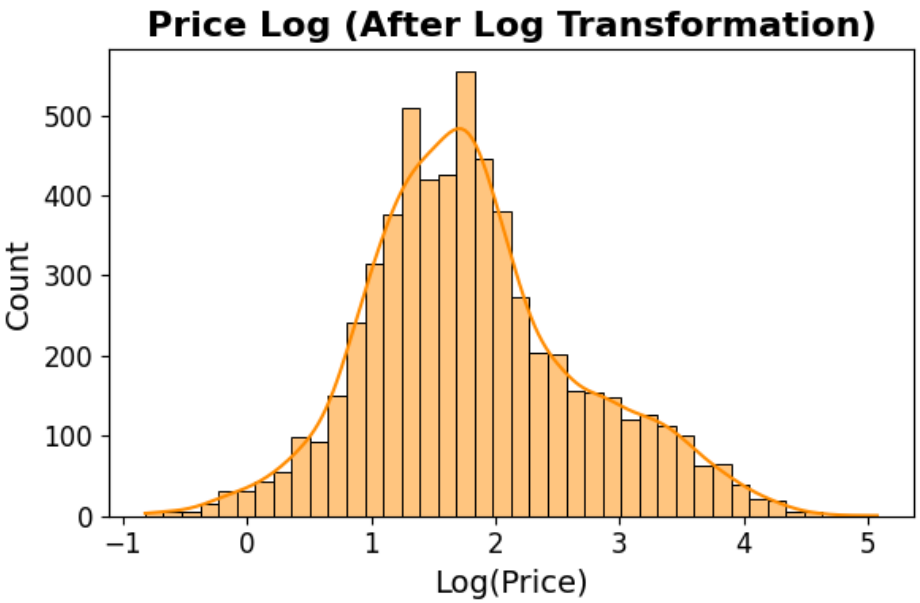


Figure A22. Car Age (Years): Boxplot and Histogram

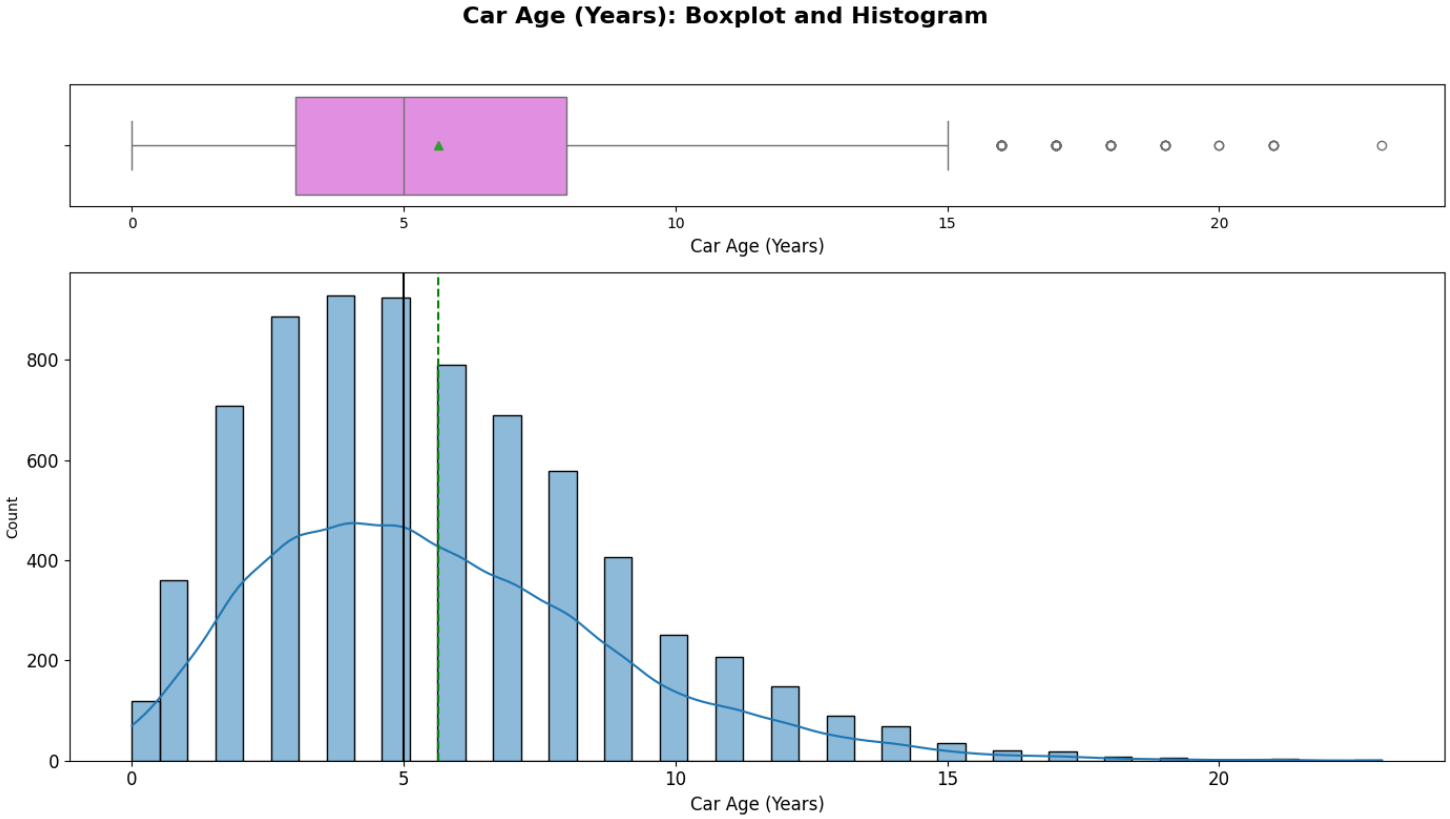


Figure A23. Depreciation Amount vs. Log(Price)

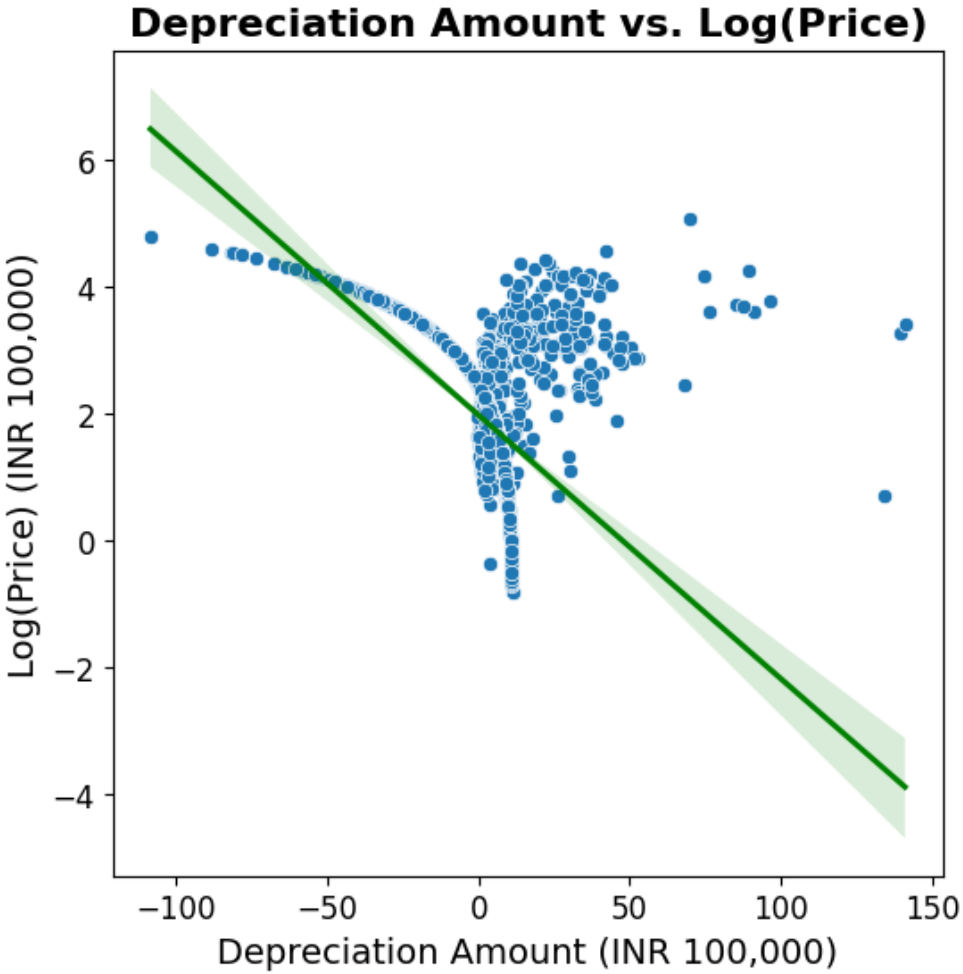


Figure A24. Average Depreciation by Brand

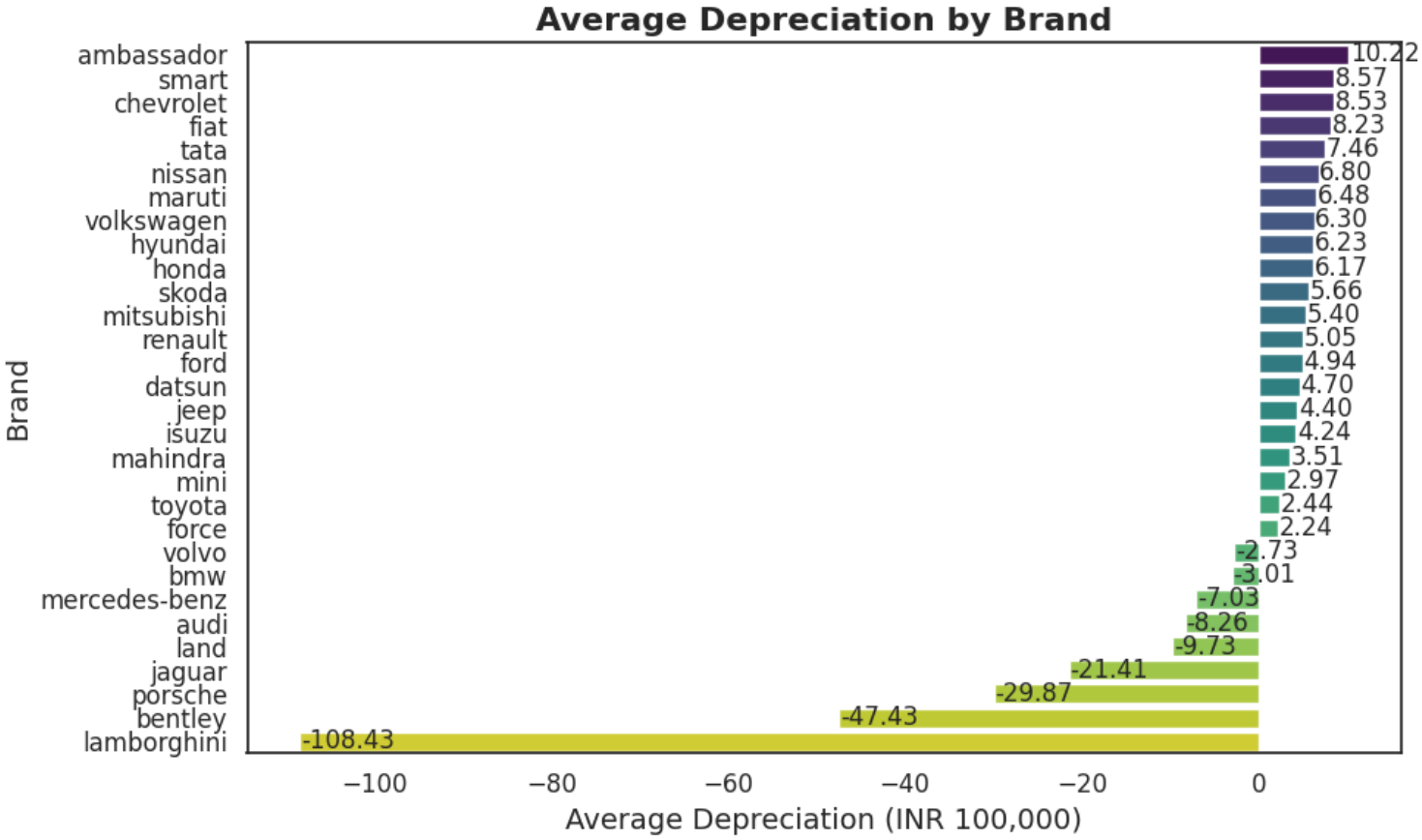


Figure A25. Training R^2 and Adjusted R^2 by Model

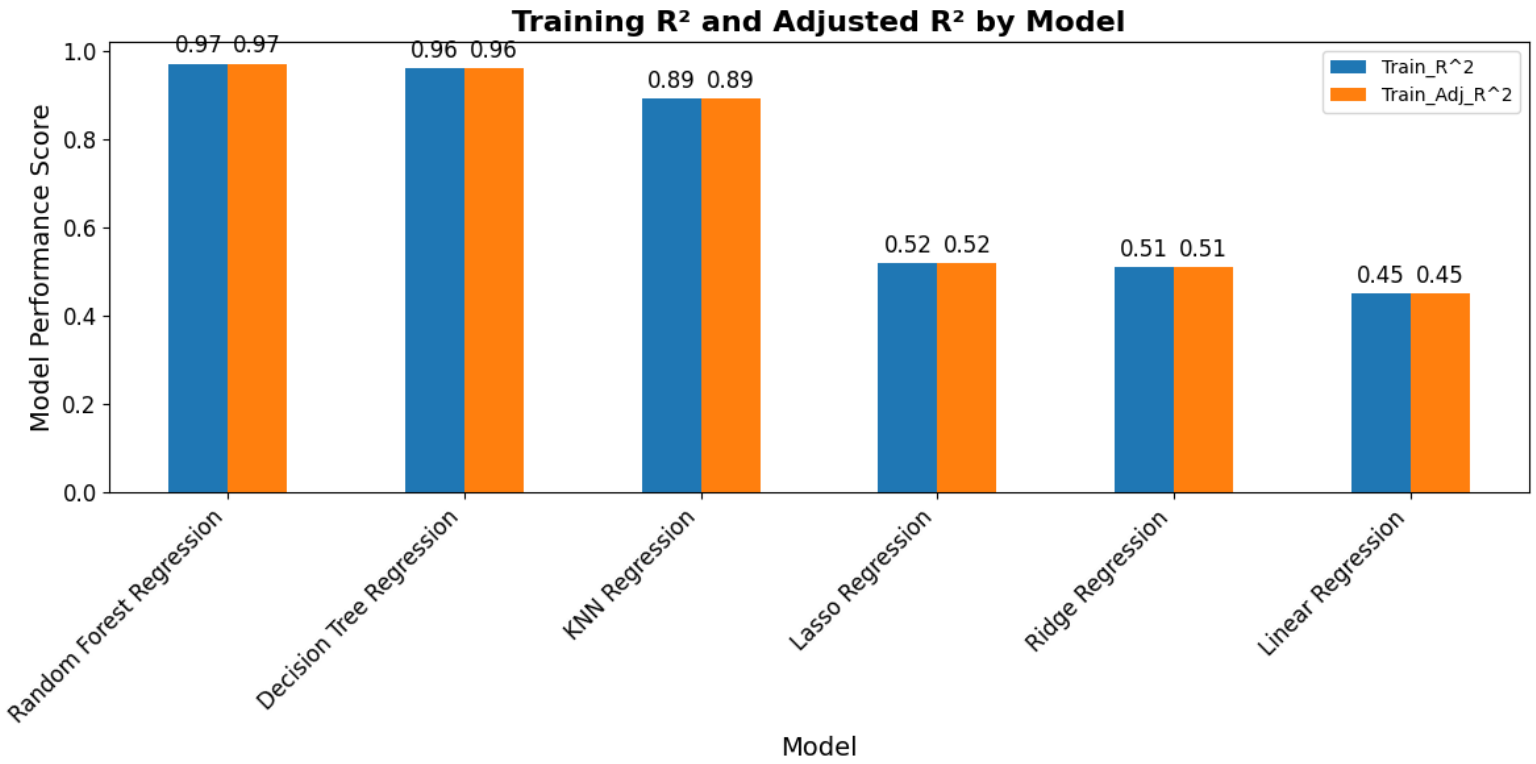


Figure A26. Test R^2 and Adjusted R^2 by Model

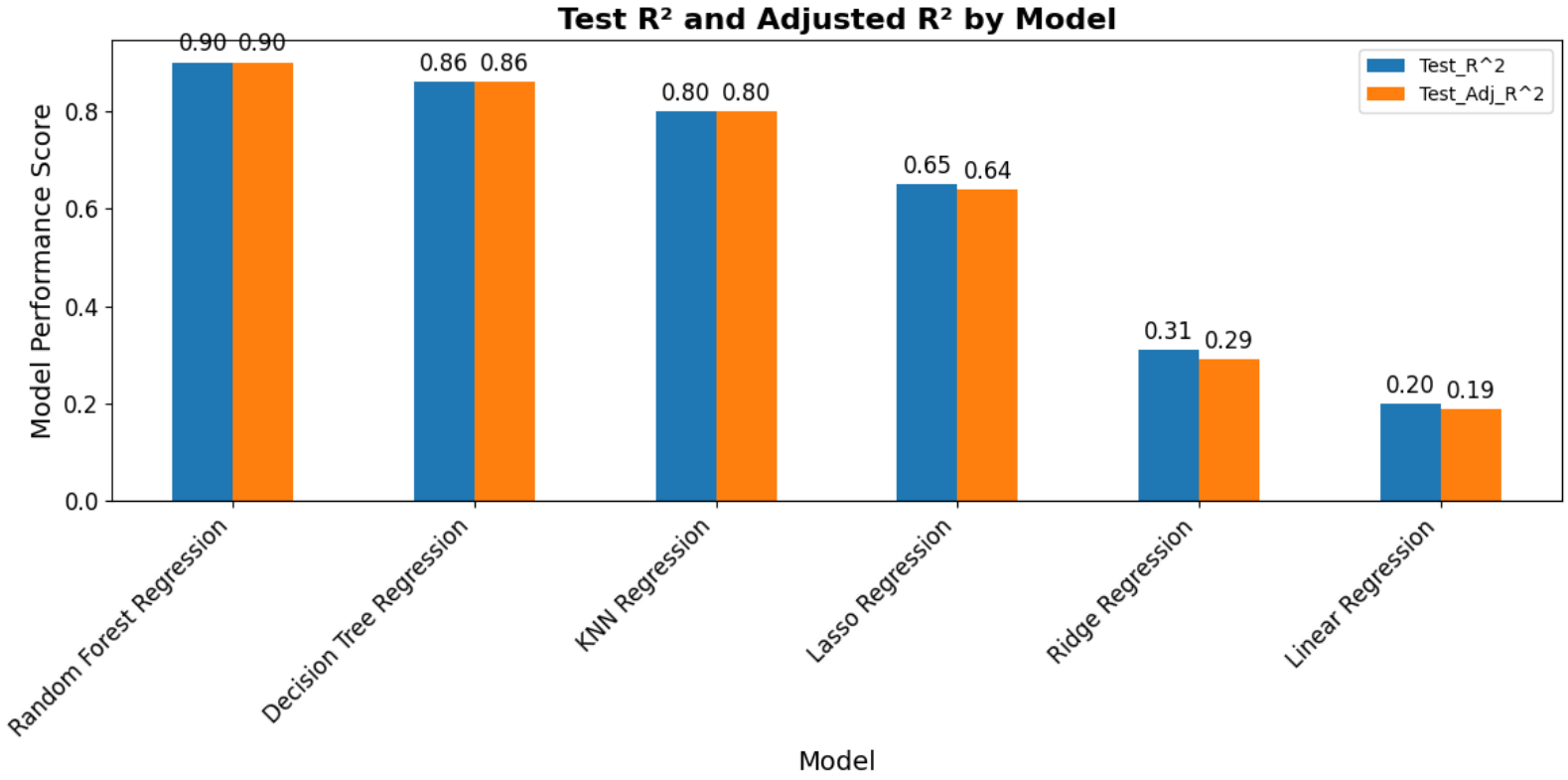


Figure A27. Training RMSE and MAE by Model

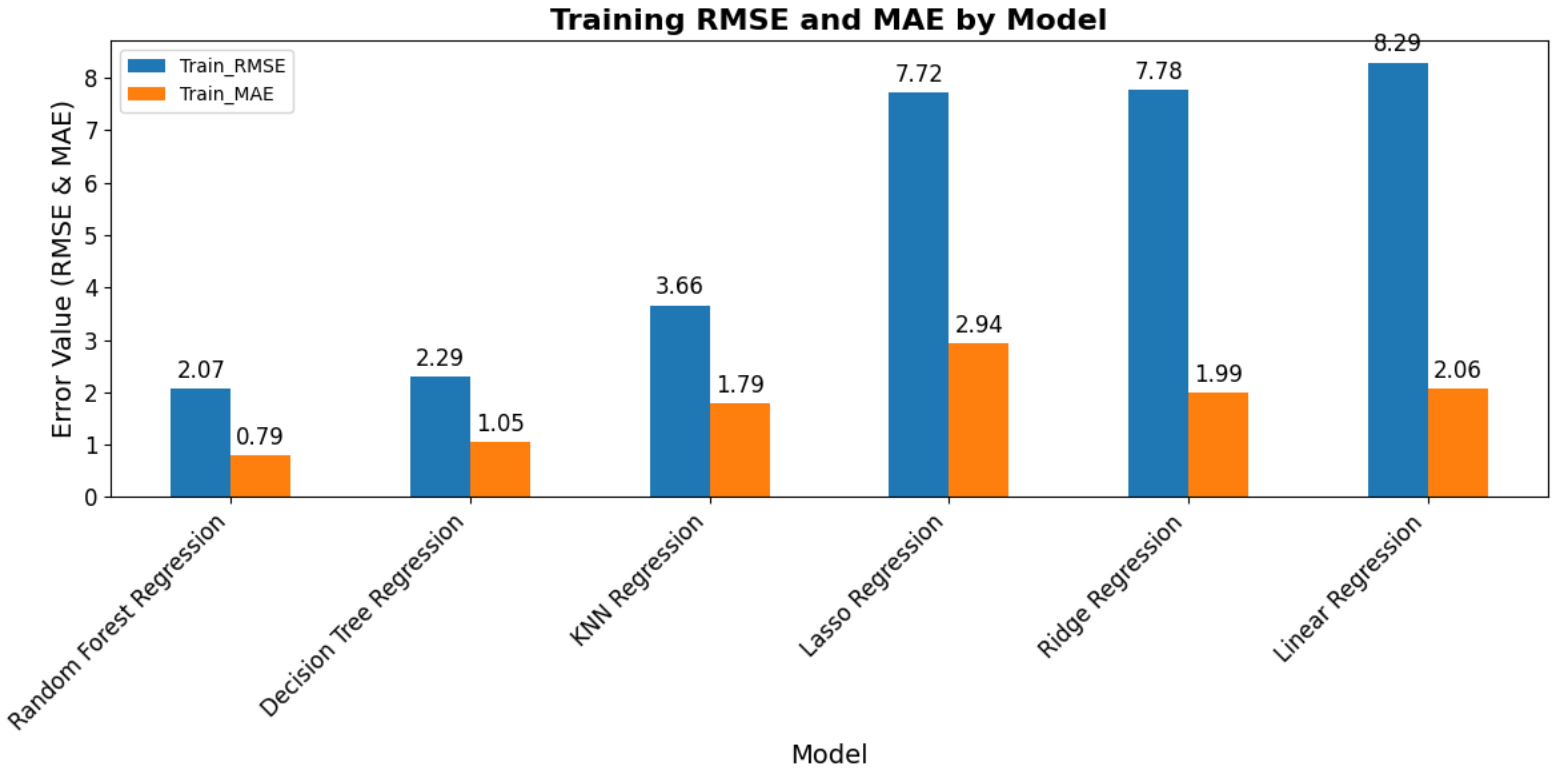


Figure A28. Test RMSE and MAE by Model

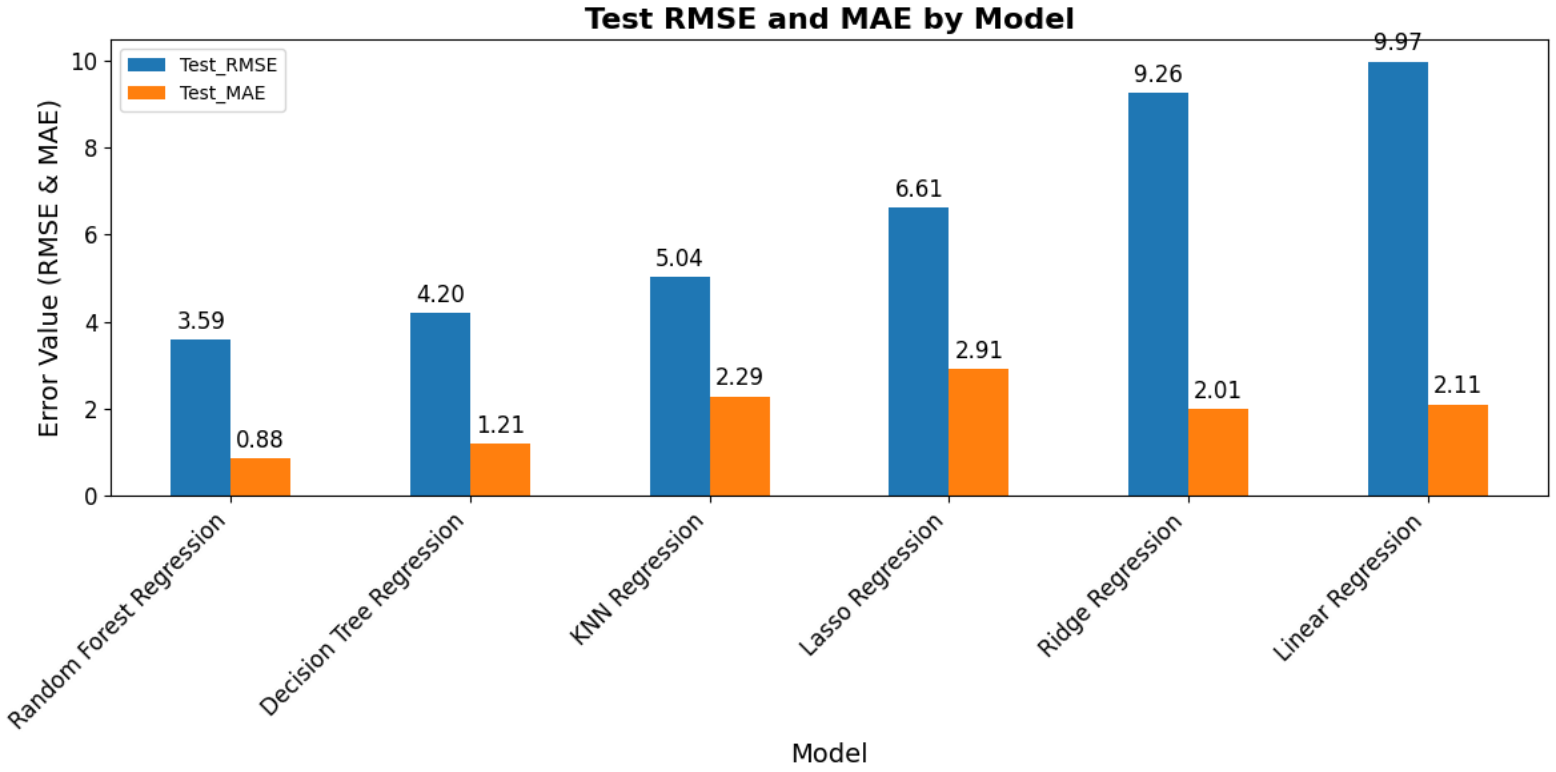


Figure A29. Training MAPE by Model

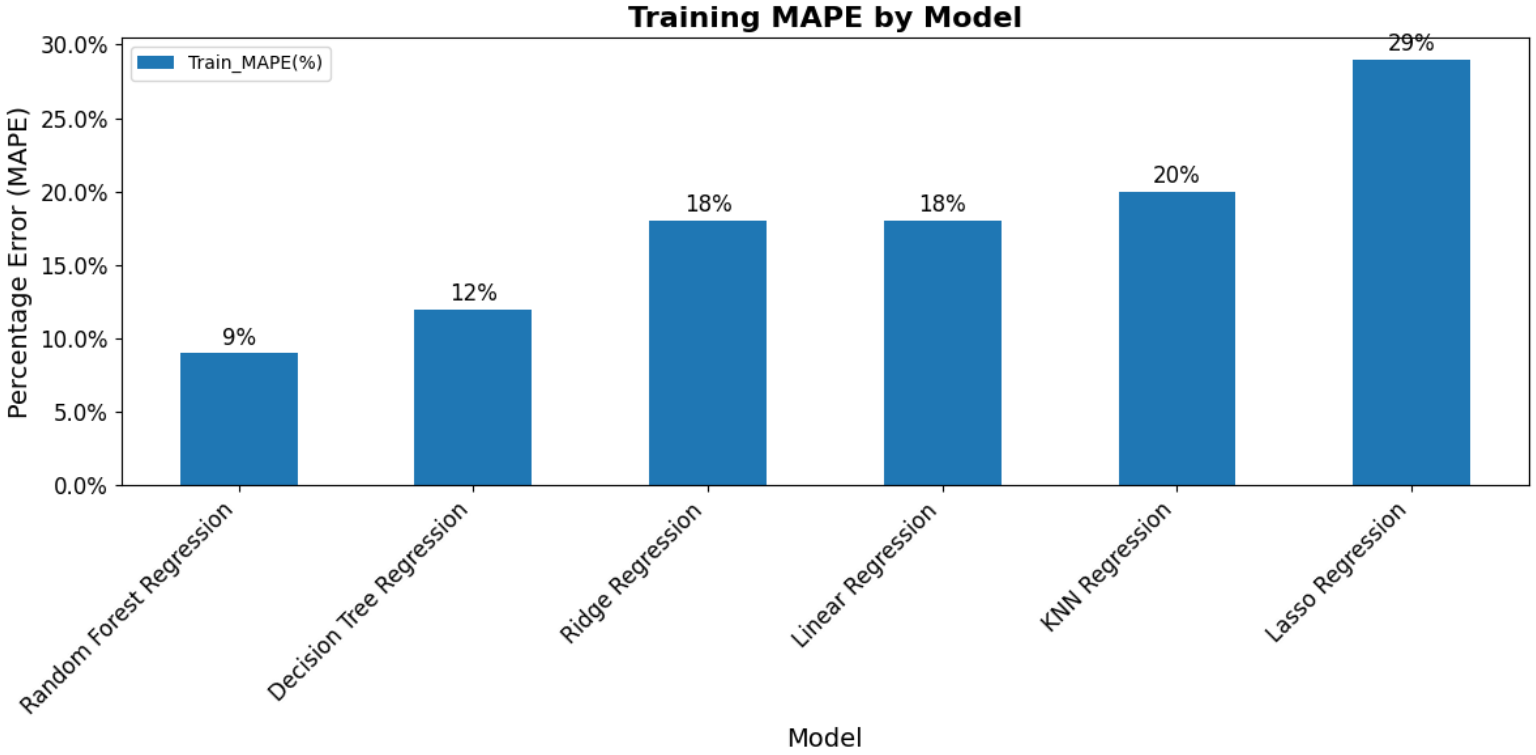


Figure A30. Test MAPE by Model

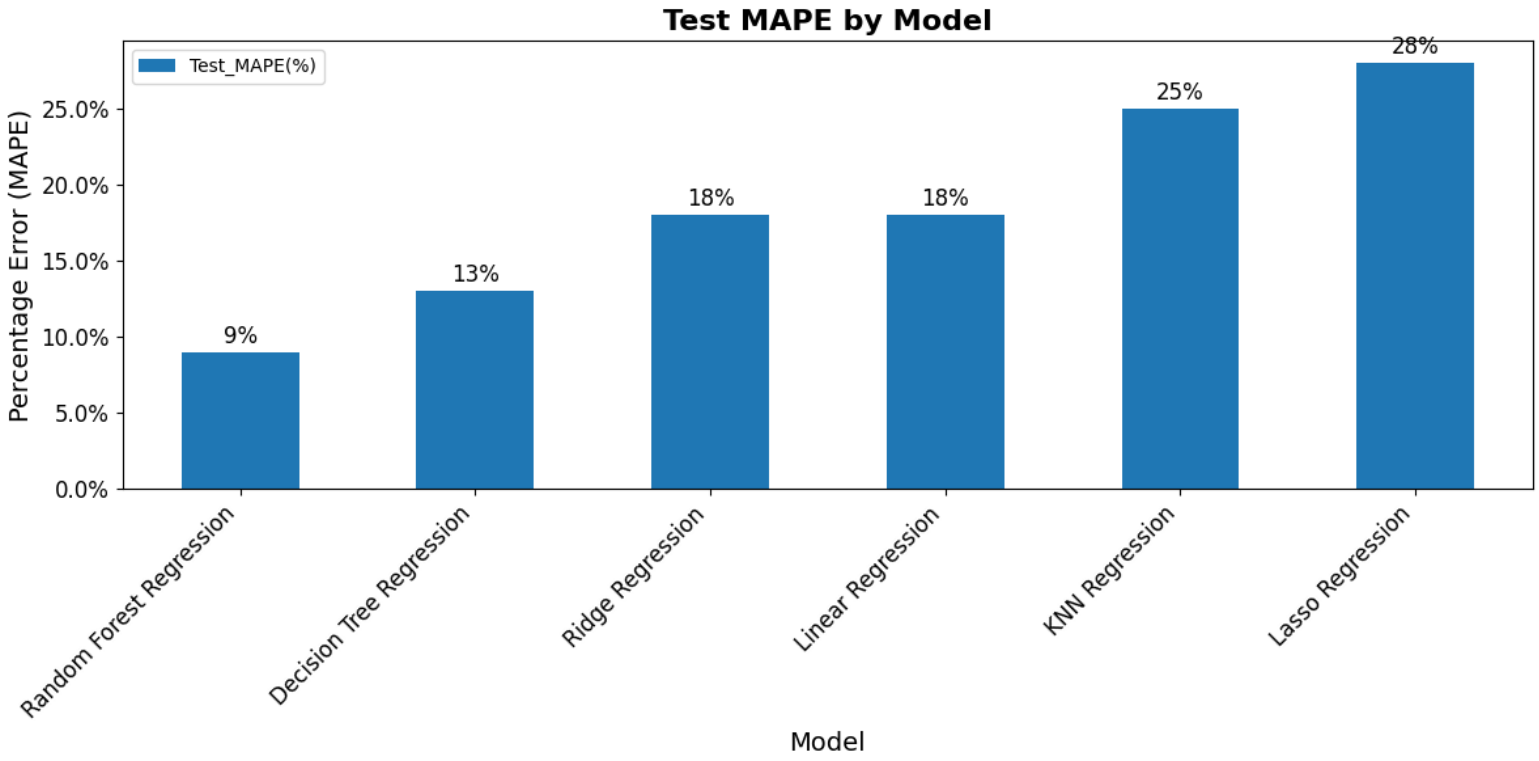


Table A1. Initial Independent Variables (Features)

```
Data columns (total 43 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Year                        4212 non-null   int64
 1   Mileage                     4212 non-null   float64
 2   Engine                      4212 non-null   float64
 3   Power                       4212 non-null   float64
 4   New_price                   4212 non-null   float64
 5   Kilometers_Driven_log       4212 non-null   float64
 6   Car_Age                     4212 non-null   int64
 7   Depreciation_Amount         4212 non-null   float64
 8   Location_Bangalore          4212 non-null   int64
 9   Location_Chennai            4212 non-null   int64
 10  Location_Coimbatore         4212 non-null   int64
 11  Location_Delhi              4212 non-null   int64
 12  Location_Hyderabad          4212 non-null   int64
 13  Location_Jaipur             4212 non-null   int64
 14  Location_Kochi              4212 non-null   int64
 15  Location_Kolkata            4212 non-null   int64
 16  Location_Mumbai             4212 non-null   int64
 17  Location_Pune               4212 non-null   int64
 18  Fuel_Type_Other_Fuel_Type   4212 non-null   int64
 19  Fuel_Type_Petrol            4212 non-null   int64
 20  Transmission_Manual         4212 non-null   int64
 21  Owner_Type_Owner_3plus      4212 non-null   int64
 22  Owner_Type_Second           4212 non-null   int64
 23  Seats_5.0                   4212 non-null   int64
 24  Seats_7.0                   4212 non-null   int64
 25  Seats_8.0                   4212 non-null   int64
 26  Seats_Rare_Seat_Values      4212 non-null   int64
 27  Brand_audi                  4212 non-null   int64
 28  Brand_bmw                   4212 non-null   int64
 29  Brand_chevrolet             4212 non-null   int64
 30  Brand_ford                  4212 non-null   int64
 31  Brand_honda                 4212 non-null   int64
 32  Brand_hyundai               4212 non-null   int64
 33  Brand_land                  4212 non-null   int64
 34  Brand_mahindra              4212 non-null   int64
 35  Brand_maruti                4212 non-null   int64
 36  Brand_mercedes-benz         4212 non-null   int64
 37  Brand_nissan                4212 non-null   int64
 38  Brand_renault               4212 non-null   int64
 39  Brand_skoda                 4212 non-null   int64
 40  Brand_tata                  4212 non-null   int64
 41  Brand_toyota                4212 non-null   int64
 42  Brand_volkswagen            4212 non-null   int64
dtypes: float64(6), int64(37)
memory usage: 1.4 MB
```

Table A2. Final Hyperparameter-Tuned Random Forest Regression Model Performance Metrics

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE (%)	Test_MAPE(%)
0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%

Figure A31. Decision Tree: Top 3 Feature Importances

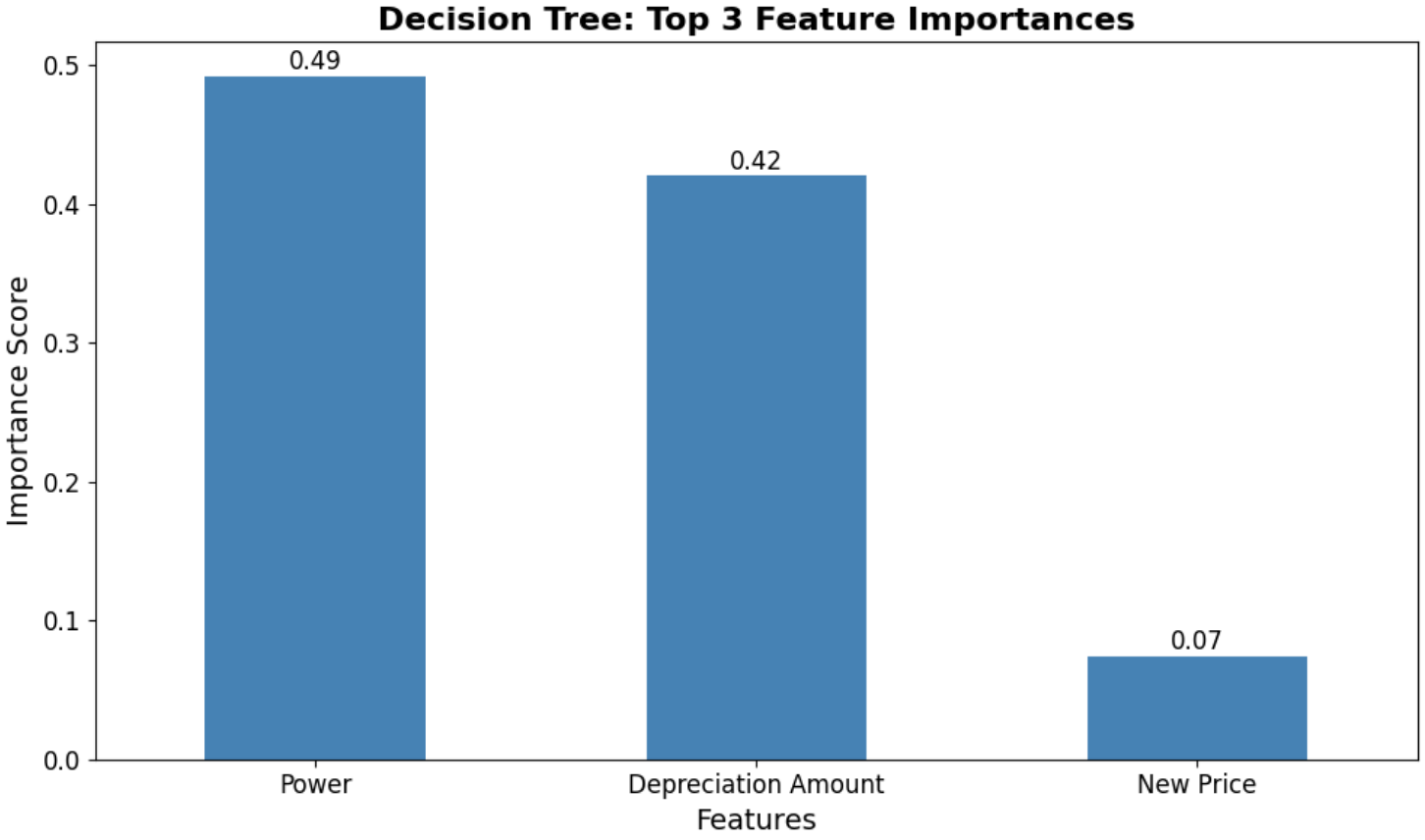


Figure A32. Random Forest: Top 3 Feature Importances

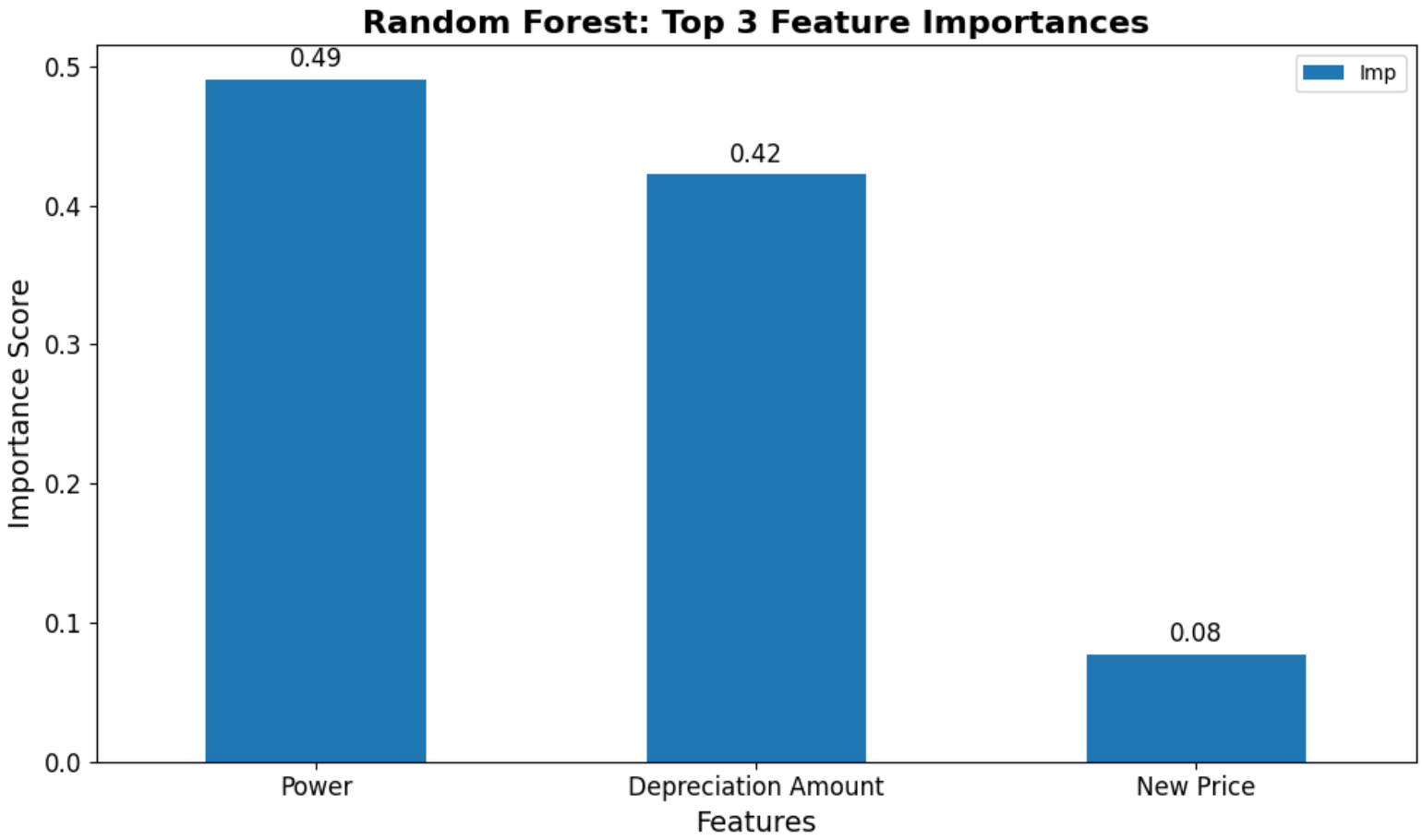


Table A3. Updated Missing Value Summary by Variable

	0
Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	83
Engine	46
Power	175
Seats	53
New_price	6246
Price	1234
Kilometers_Driven_log	0
Price_log	1234
Brand	0
Model	0
Car_Age	0

Table A4. Performance Metrics of Final Regression Models

	Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
Linear Regression	0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	18%	18%
Ridge Regression	0.51	0.31	0.51	0.29	7.78	9.26	1.99	2.01	18%	18%
Lasso Regression	0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	29%	28%
Decision Tree Regression	0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	12%	13%
Random Forest Regression	0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%
KNN Regression	0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	20%	25%

Table A5. Linear Regression Model Comparison – Baseline vs. Final

Baseline (originally with too much noise)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.50	0.29	7.79	9.27	1.99	2.01	18%	18%

Final (with multicollinearity and p-values addressed)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	18%	18%

Table A6. Ridge Regression Model Comparison – Baseline vs. Final

Baseline (default parameters)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.51	0.29	7.78	9.26	1.99	2.01	18%	18%

Final (addressed INF multicollinearity)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.50	0.29	7.78	9.25	1.99	2.01	18%	18%

Table A7. Lasso Regression Model Comparison – Baseline vs. Final

Baseline (fits worse than a horizontal line at the mean of the target)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
-0.08	-0.10	-0.09	-0.13	11.63	11.69	5.98	6.21	80%	80%

Final (with alpha 0.1 tuned)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	29%	28%

Table A8. Decision Tree Regression Model Comparison – Baseline vs. Final

Baseline (severely overfitted, memorizes training data, and lacks generalization)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
1.0	0.91	1.0	0.91	0.0	3.43	0.0	0.38	0%	2%

Final (hyperparameter tuned, max_depth = 5, min_samples_leaf = 5, random _state = 1)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	12%	13%

Table A9. Random Forest Regression Model Comparison – Baseline vs. Final

Baseline (too tightly fitted the training data, not learning a stable underlying pattern)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
1.0	0.92	1.0	0.92	0.67	3.13	0.09	0.26	1%	1%

Final (hyperparameter tuned, max_depth = 5, min_samples_leaf = 1, criterion = squared error, random_state = 1)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%

Table A10. KNN Regression Model Comparison – Baseline vs. Final

Baseline (with k = 10 neighbors)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.85	0.75	0.85	0.74	4.32	5.53	2.18	2.52	25%	27%

Final (originally baseline with default parameters)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	20%	25%